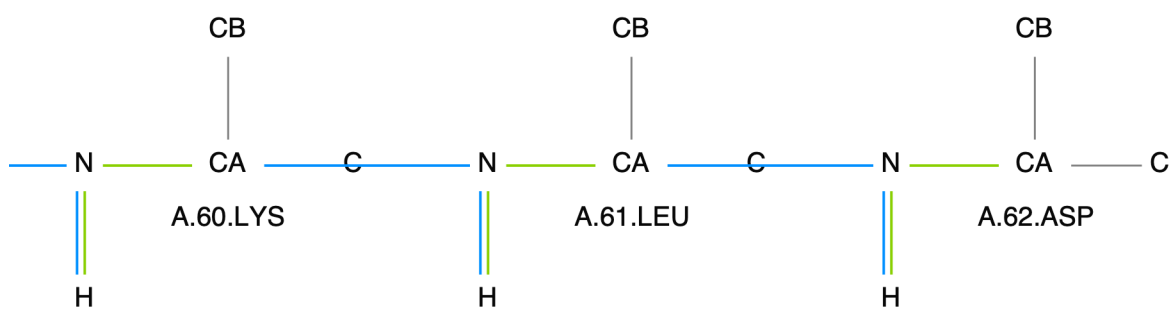
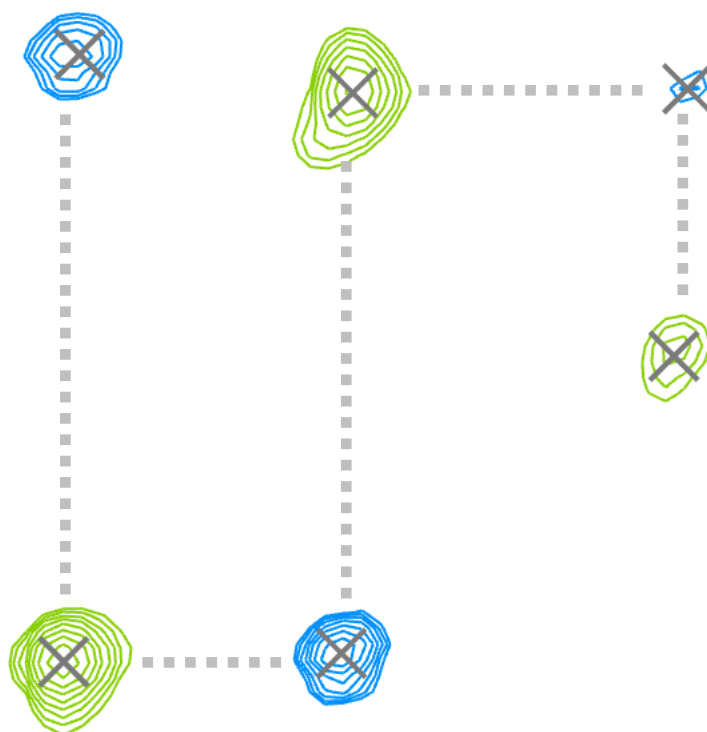


SH3 ^1H Detection Assignment Tutorial

Introduction

This tutorial is designed to guide you through a solid-state NMR sequential triple resonance backbone assignment using Ccpnmr AnalysisAssign Version 3.3. It is not intended to teach any theoretical aspects of NMR and protein assignment.

In this tutorial you will use spectra recorded on the SH3 domain of chicken α -spectrin which was ^2H , ^{13}C , ^{15}N labelled and back-exchanged into 100% $^1\text{H}_2\text{O}$. The spectra were recorded at 100 kHz MAS on an 800 MHz spectrometer. The spectra used, are a 2D hNH and the following set of 3D spectra: hCANH, hcoCAcoNH, hcaCBcaNH, hCONH and hCOcaNH. We are grateful to Prof. Bernd Reif for providing us with these spectra.

The tutorial uses projects provided in the **CcpnSH31HDetAssignTutorial** directory.

Please note that the images shown are only representative and you may encounter minor differences to your setup.

Contents:

1. Project Setup
2. Peak Picking Spectra
3. Sequential backbone assignment
4. Other Useful Tools
5. Reference Information

Start CcpNmr Analysis V3

Apple users by double clicking the icon
CcpNmrAnalysis



Linux users by using the terminal
command: *bin/assign*

Windows users by double-clicking on the
assign.bat file

Reminder: basic operations in CcpNmr Analysis

Sidebar

All data contained in a project, such as spectra and peak lists are located in the sidebar. **Double-clicking** on an item will open its properties popup. **Dragging** an item into the **Drop Area** will open it as a module (Spectrum Display, Table etc.)

Spectrum Display

A Spectrum Display can contain multiple overlaid spectra which share the same axes. To show/hide a single spectrum, click on its spectrum toolbar button. If you close a display, you can open a spectrum by **dragging and dropping** it into the drop area from the sidebar or by **right-clicking** on a sidebar item and selecting **Open as module**. You can also add additional spectra to a spectrum display module or drag several spectra into the drop area together to open them simultaneously.

Mouse

- Pan -> **Left-drag** in display
- Zoom in/out -> **Scroll wheel** in display
- Context menu -> **Right-click**
- Select a peak -> **Left-click** on a peak symbol "X"
- Move a peak -> select first, then **right-click and drag**

Two-Letter Shortcuts

Press the first letter on your keyboard e.g., **M**, followed by the second letter, e.g., **K** (case insensitive). Press **Esc** to cancel the first letter.

For more commands and operations

Main Menu → Help → Tutorials → Beginners Tutorial

OR

Main Menu → Help → Show Shortcuts

Main Steps

• Open project

- Drag in **SH31HdetSpectra.ccpn**

1

• Project Setup:

- Spectrum > Set Experiment Types (ET)
- Sidebar > Chain > <New Chain> using protein name and sequence

• Pick and Assign 3D Spectra:

- Assign > Pick and Assign (PA) using Settings:
 - Tick **Automatic C/CA/CB NmrAtom assignment (Cas -ve)**
 - For **Pick Peaks** select **Display** and all 3Ds from drop-down
- Pick a peak in the hNH spectrum
- Assign > Set up NmrResidues (SN)
- **Double-click** on an NmrResidue in Pick & Assign table to navigate
- Pick 3D peaks or use **Pick and Assign**
- **Click Assign Selected** to assign H, N and C atom types
- Correct carbon atom assignments, if necessary, using Assignment > NmrAtom Assigner (AN)

2

• Open project SH31HdetPeakPickedAndAssigned.ccpn

• Sequential Assignment / Linking NmrResidues:

- Assign > Backbone Assignment (BB) using Settings:
 - Match CA and C atoms. Set **Match** and **Search** modules to top two 3D Spectrum Displays, **Add Display Group** for lower two 3Ds
- Double-click on NmrResidue @68 in BB Assignment table
- Drag NmrAtom of matching strip onto orange arrow in Search Spectrum Display:

GS:3D_HCN_2.1	- C2	GS:3D_HCN.1	GS:3D_HCN.2	GS:3D_HCN.3
NR:@-.@68	←	NR:@-.@123.	NR:@-.@153.	NR:@-.@77.
N:116.580[1]	- 43	[i+1] [99%]	[i+1] [91%]	[i+1] [56%]

- Continue linking in i+1 direction
- For i-1 direction double click @69-1 and drag onto left hand arrow

• Make Sequence Specific Assignment:

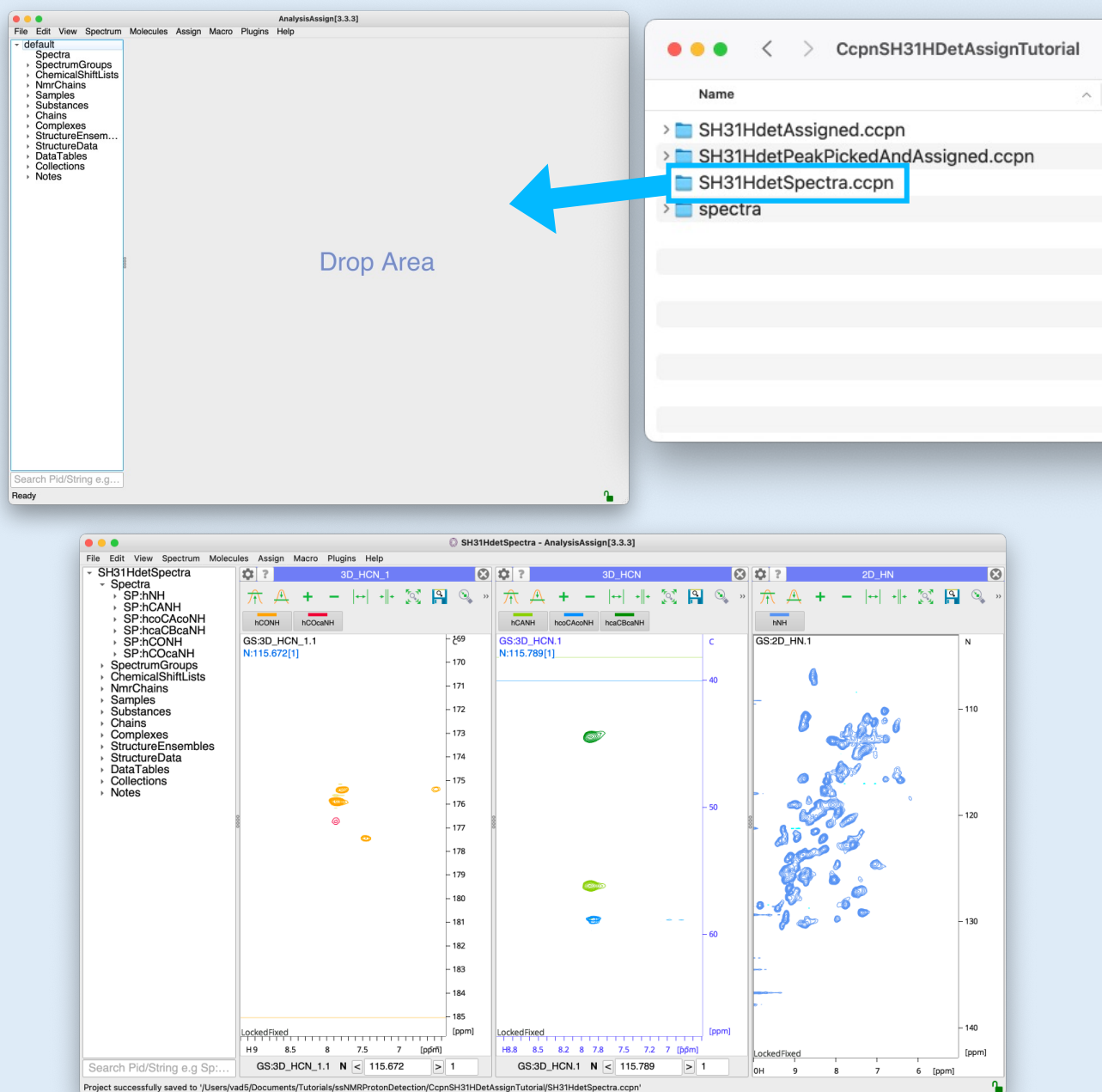
- In Sequence Graph: drag an NmrResidue onto matching Residue in Sequence (in lower part of module) to assign the whole stretch

• Other Useful Tools:

- Check/correct assignments using Peak Assigner (AP) Assignment Inspector (AI), Sequence Graph (SG), Reference Chemical Shift (RC), Residue Information (RI)

4

Open the project **SH31HdetSpectra.ccpn**



1A Drag & drop SH31HdetSpectra.ccpn into the sidebar or drop area.

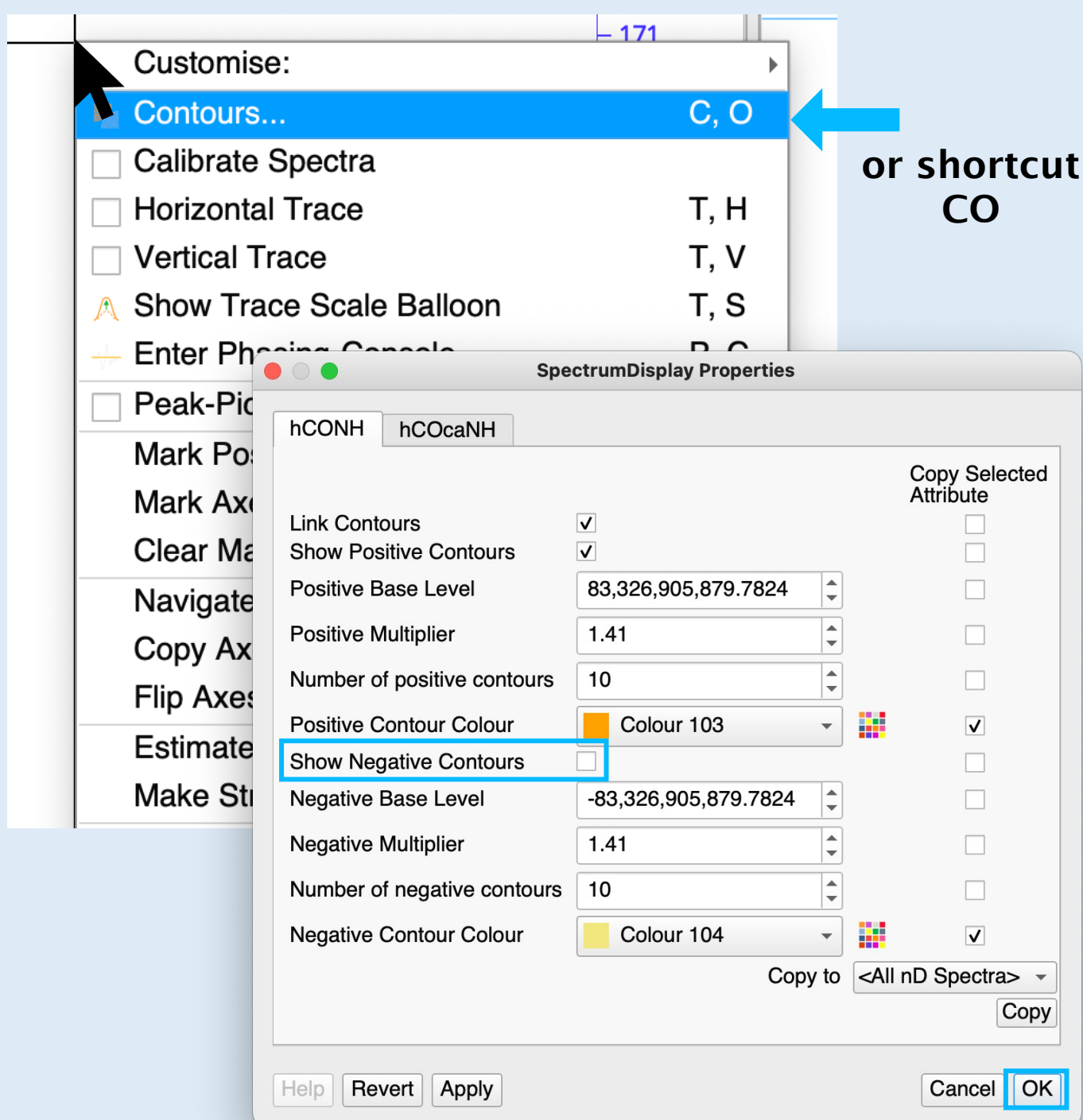
CcpNmr projects are saved as directories with the **.ccpn**.

- Find the project directory **SH31HdetSpectra.ccpn** in the **CcpnSH31HDetAssignTutorial** directory and drag & drop it into the program.

The project will be loaded in your program window.

You will see six spectra, displayed in three SpectrumDisplays as:

- hNH (light blue, 2D_HN)
- hCANH (light green, 3D_HCN)
- hcoCAcoNH (blue, 3D_HCN)
- hcaCBcaNH (dark green, 3D_HCN)
- hCONH (orange, 3D_HCN_1)
- hCOcaNH (red, 3D_HCN_1)

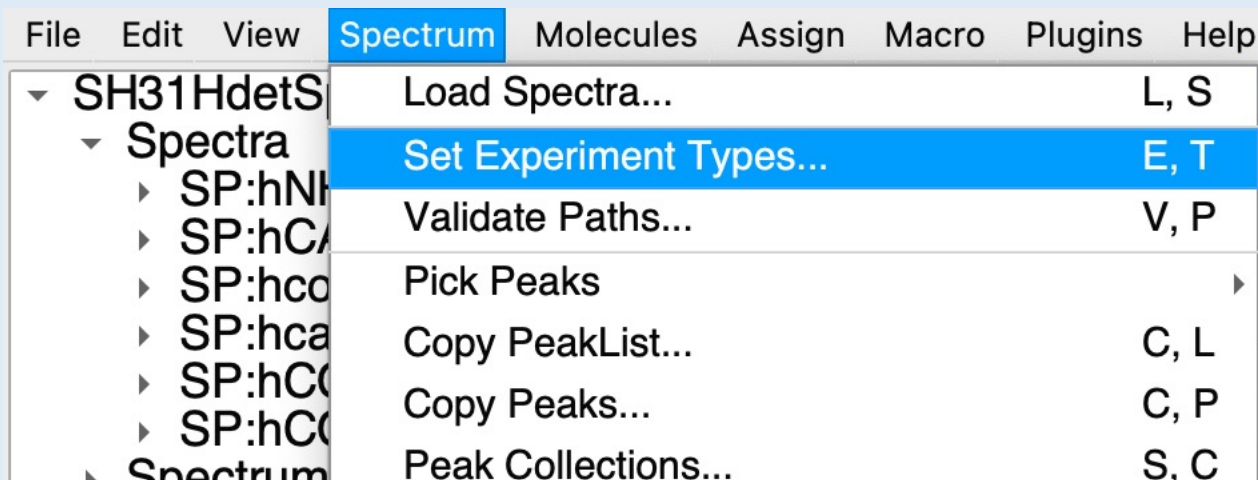


1_B Set Spectrum Properties: Contours.

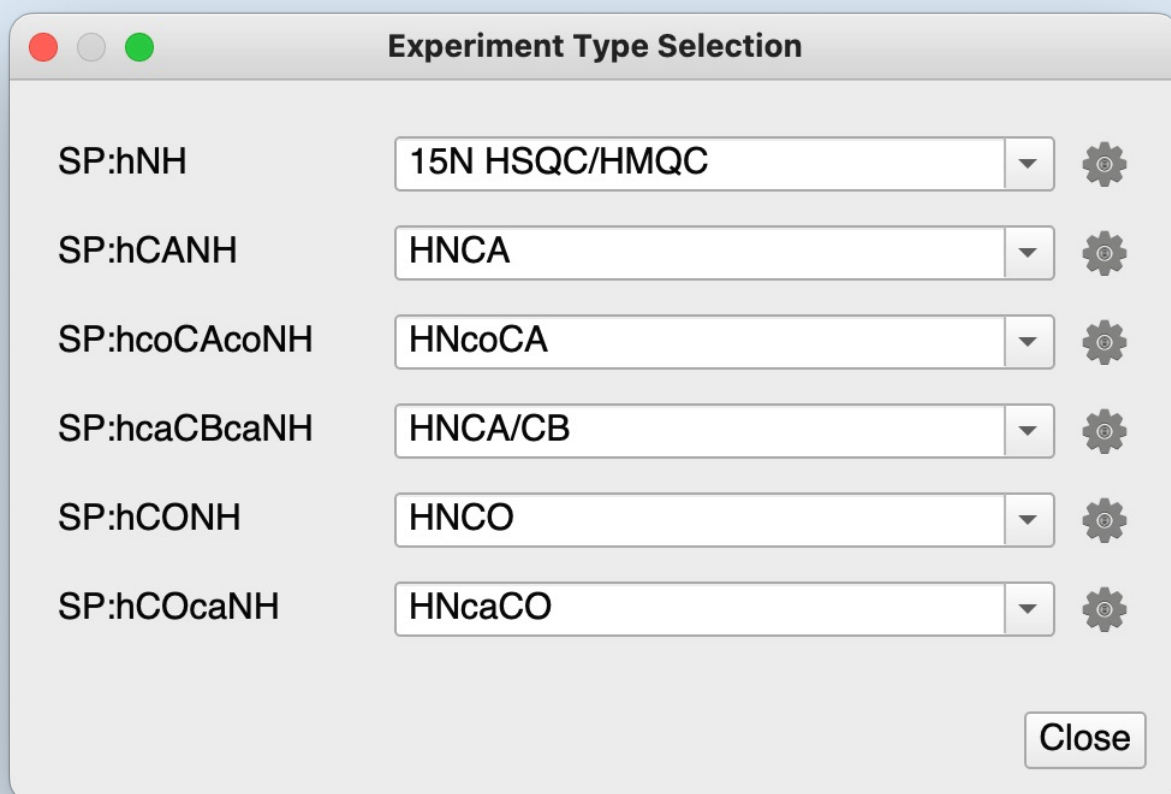
Negative contours can be useful to get a sense of the noise level in a spectrum, but to avoid picking negative peaks by mistake, it is best to turn them off:

- **right-click** on each spectrum display and click **Contours...**
- uncheck **Show Negative Contours**

You can also perform these actions for each individual Spectrum in the **Spectrum Properties** popup; open them by **double-clicking** on the spectra in the sidebar.



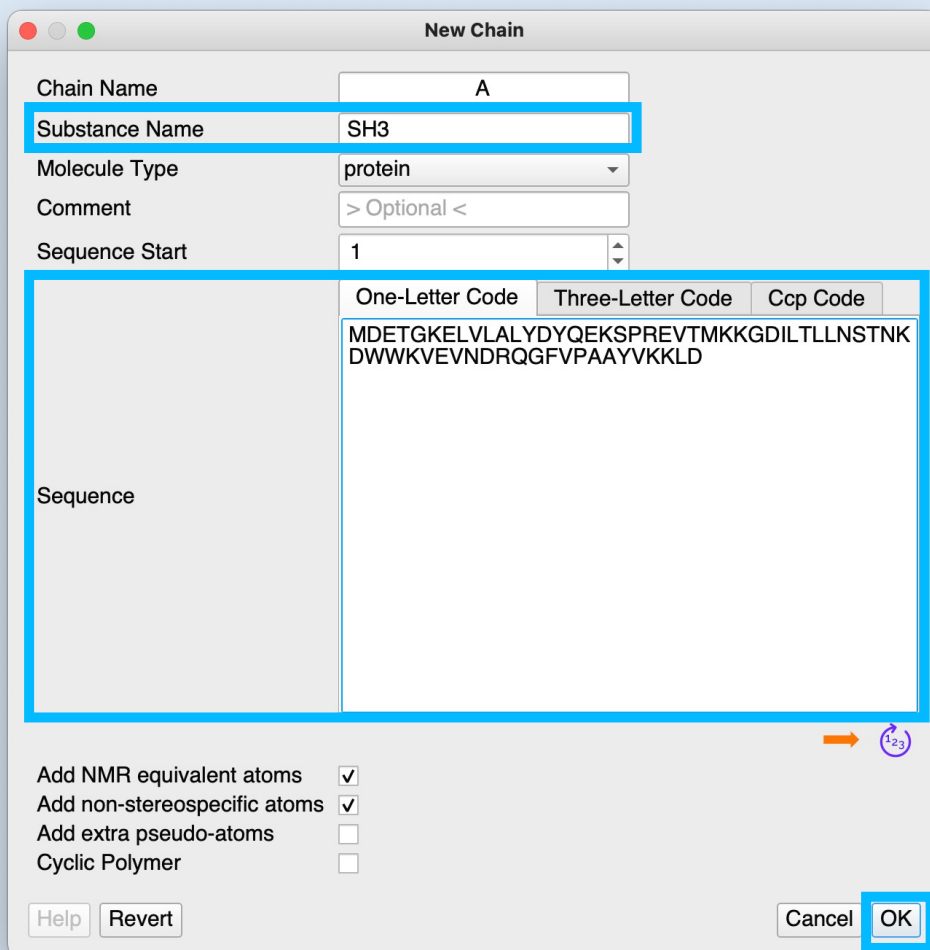
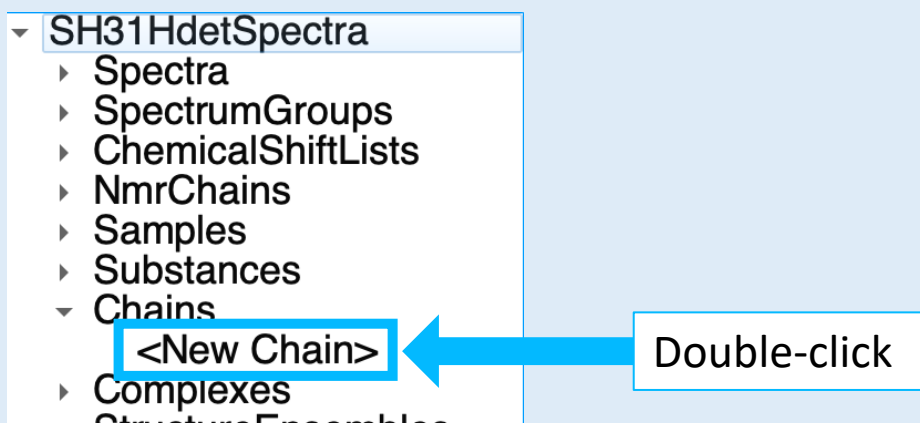
or shortcut ET



1C Set Experiment types

It is possible to tell the program what kind of experiment each spectrum is. Although this is not strictly necessary in order to assign you spectra, it will allow you to make use of several helpful features later on. Simply choose the solution NMR equivalents of the solid-state spectra used here.

- Go to **Main Menu** → **Spectrum** → **Set Experiment Types...** or shortcut **ET**
- Set the Experiment Types from the drop-down menus as shown above and then click **Close**.



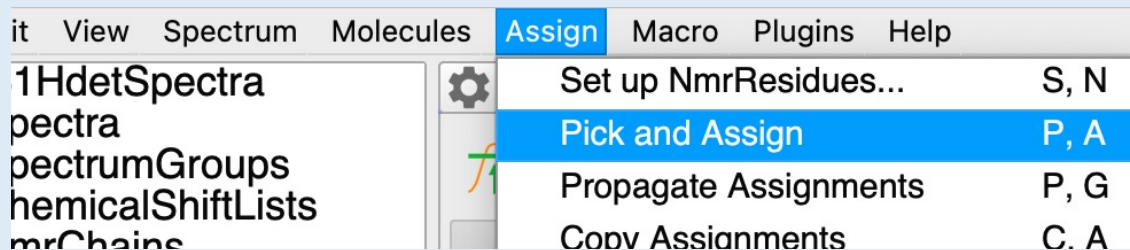
1D Create Chain

- Go to **Sidebar** → **Chains** → <NewChain>

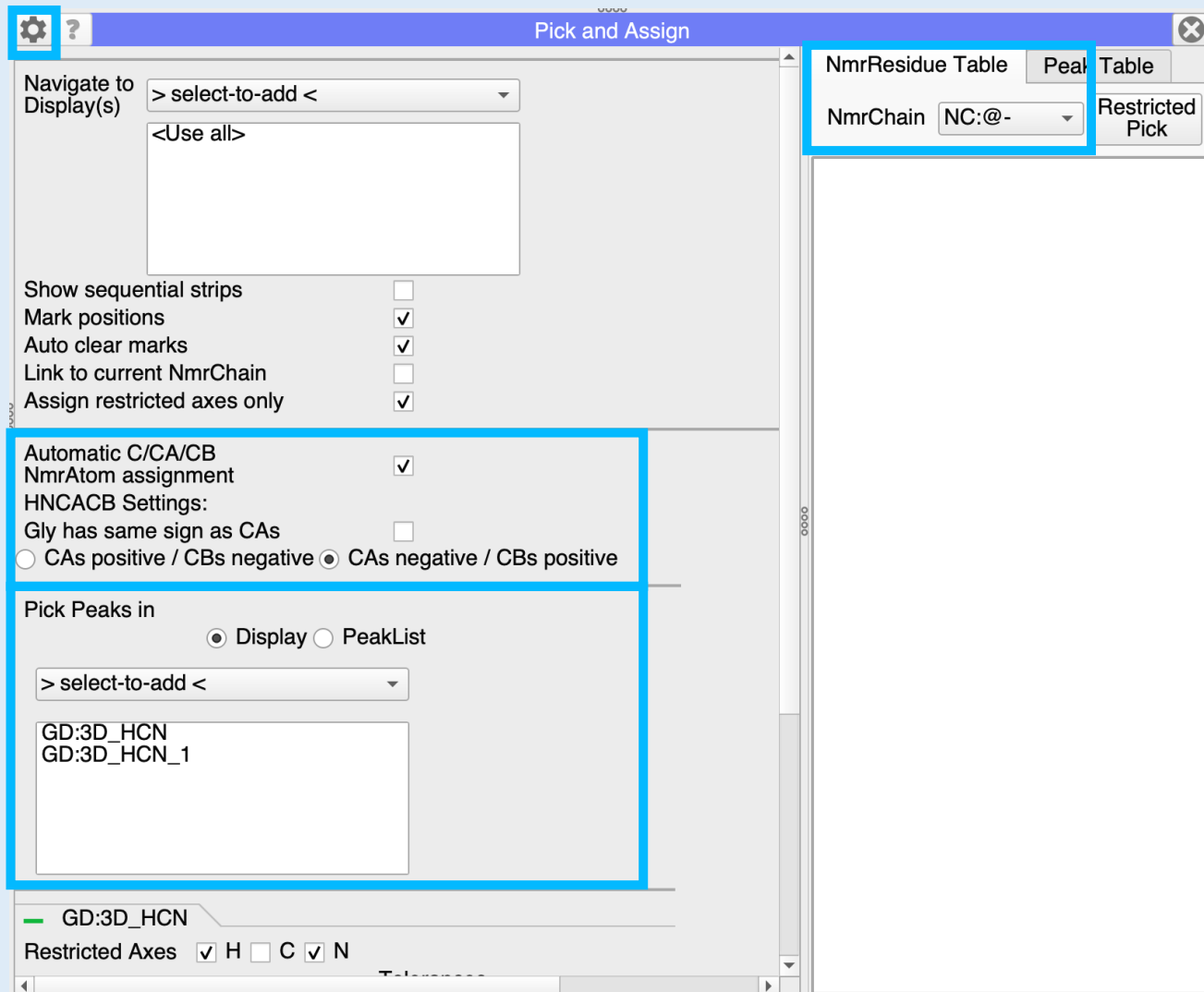
Use the default settings with

- **Substance Name:** SH3
- **Sequence:** (copy and paste)
MDETGKELVLALYDYQEKS PREVTMKKGDILTLLNSTNK DWWKVEVNDRQGFVPAAYV
KKLD
- Click **OK**

You can also use **Main Menu** → **Molecule** → **Generate Chain** or drop a FASTA formatted file into the project (e.g. **sh3_1m8m.fasta** in the tutorial data folder).



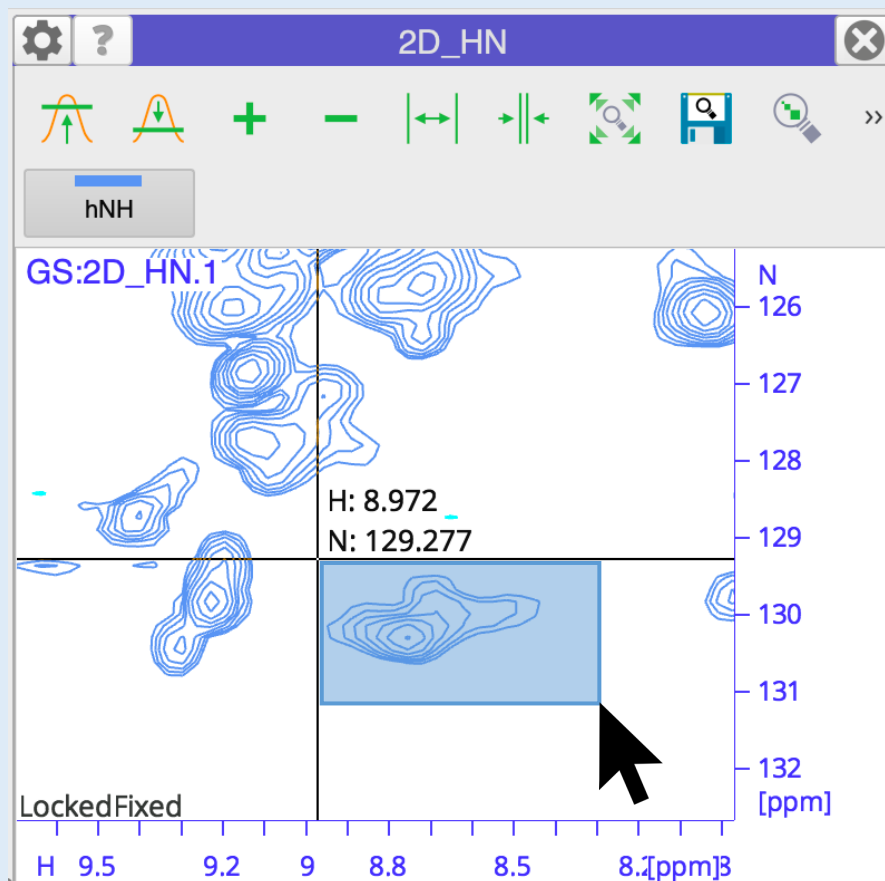
or shortcut PA



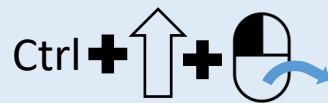
2A Pick and Assign Peaks Setup

We will pick the 3D spectra based on the hNH spectrum using the **Pick and Assign** module:

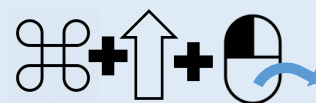
- Go to **Main Menu** → **Assign** → **Pick and Assign**, or shortcut **PA**.
- Open the Settings (gear icon ) and select:
 - **Automatic C/CA/CB NmrAtom assignment** (using the options shown above)
 - Pick Peaks in **Display**
 - **3D_HCN** and **3D_HC_1**
(if required, remove <Use All> with **right-click / Remove**)
 - Leave the remaining options as default and close the settings
- Ensure the selected NmrChain is **NC:@-**.



Linux / Windows:



Mac:



2B Picking peaks in the hNH spectrum

The relatively high degree of overlap in the hNH spectrum, means that it is better to pick peaks individually rather than in one go.

Picking an obvious maximum:

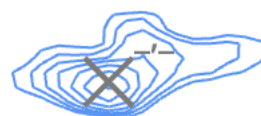
- Click on the spectrum display, hold down **Ctrl** (or **Cmd** for Mac) + **Shift** and **left-drag** the mouse to create a blue peak picking box.

Placing a peak in an overlapped area:

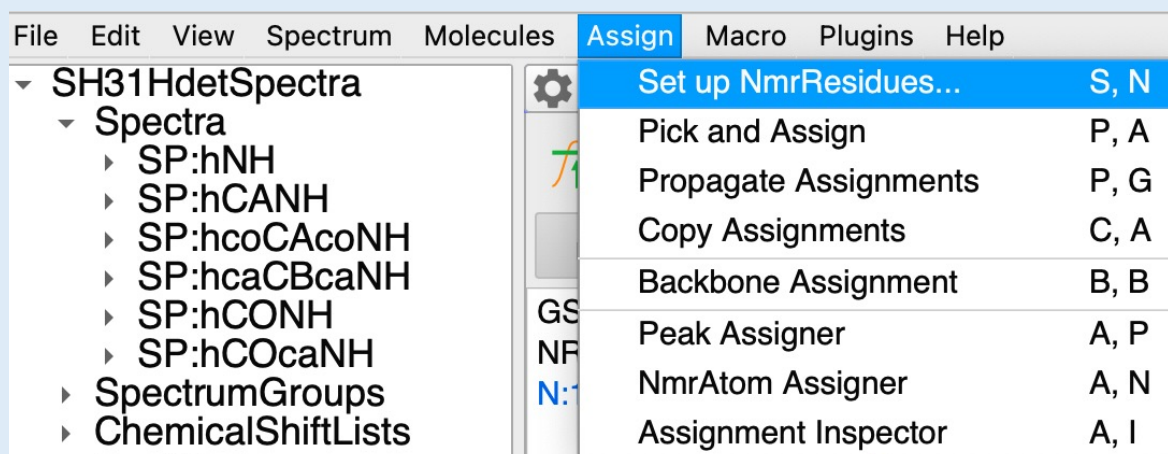
- Hold down **Ctrl** (or **Cmd** for Mac) + **Shift** and **click** the mouse at the position where you would like to place the peak.

A cross marks where a peak has been picked.

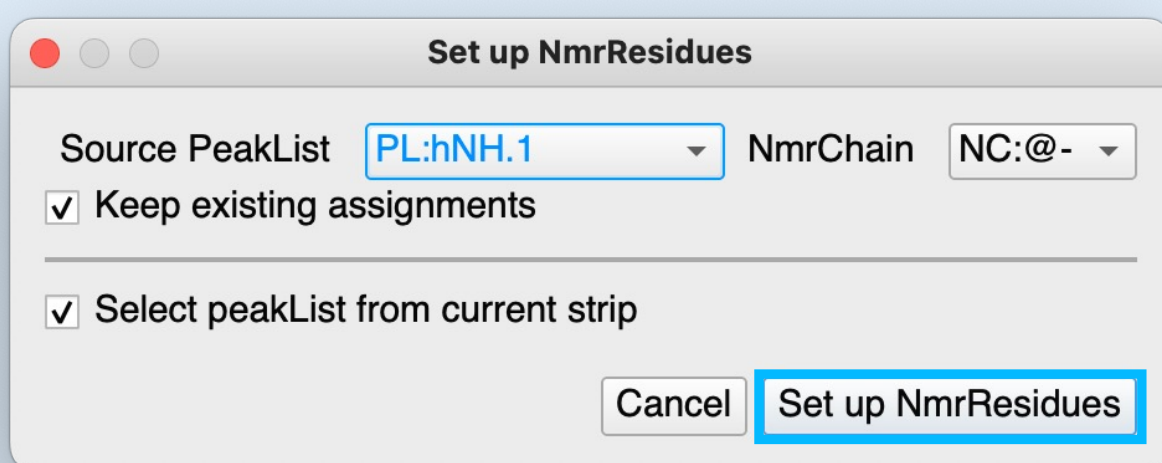
The hyphens separated by a comma indicate that the dimensions of these peaks are unassigned.



In principle you could pick the whole hNH spectrum before continuing, but in practice it is probably better to pick the 2D spectrum at the same time as you pick the 3D spectra, as the 3Ds will give you a better indication of where you need to pick/place your peaks in the 2D.



or Shortcut **SN**



2C Setup NmrResidues

Once you have picked your first peak in the hNH spectrum, you should use **Set up NmrResidues** to give your peak an initial random residue number by which to identify it and link it to peaks in other spectra as well as labeling the atoms.

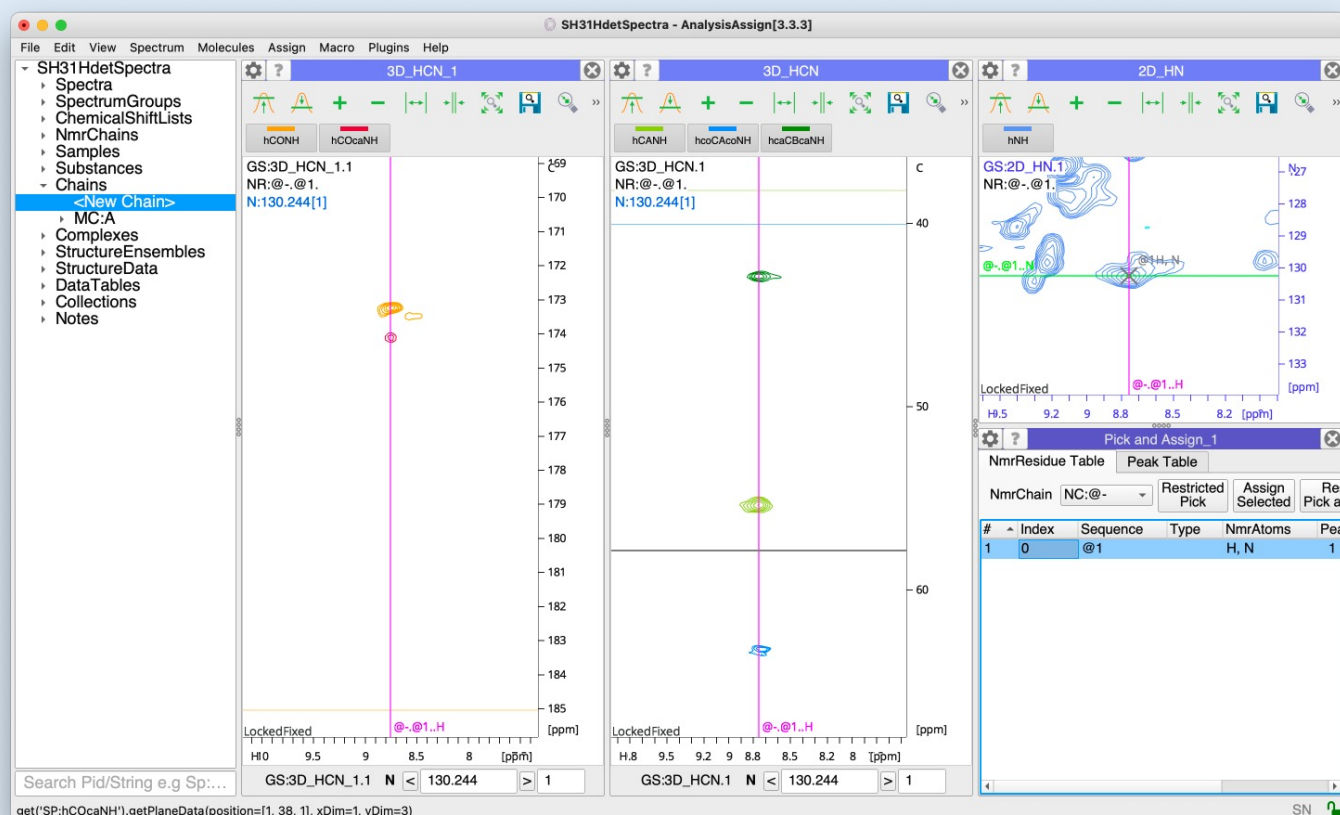
- Go to **Main Menu** → **Assign** → **Setup NmrResidues** (or use shortcut **SN**)
- Source PeakList: **PL:hNH.1**
- NmrChain: **NC:@-**
- Click **Set up NmrResidues**

Your peak will now have a label such as **@1H,N**.



Use the shortcut **PL** to toggle between different ways to display the peak labels or go to the **SpectrumDisplay Settings** (click on the gear icon in the top left hand corner) to set the way you wish to view your PeakLabels. There are also further settings if you **double-click** on the PeakList in the sidebar.

Arrange your modules with the **Pick and Assign** module below or above the **2D_HN** (hNH containing) module:



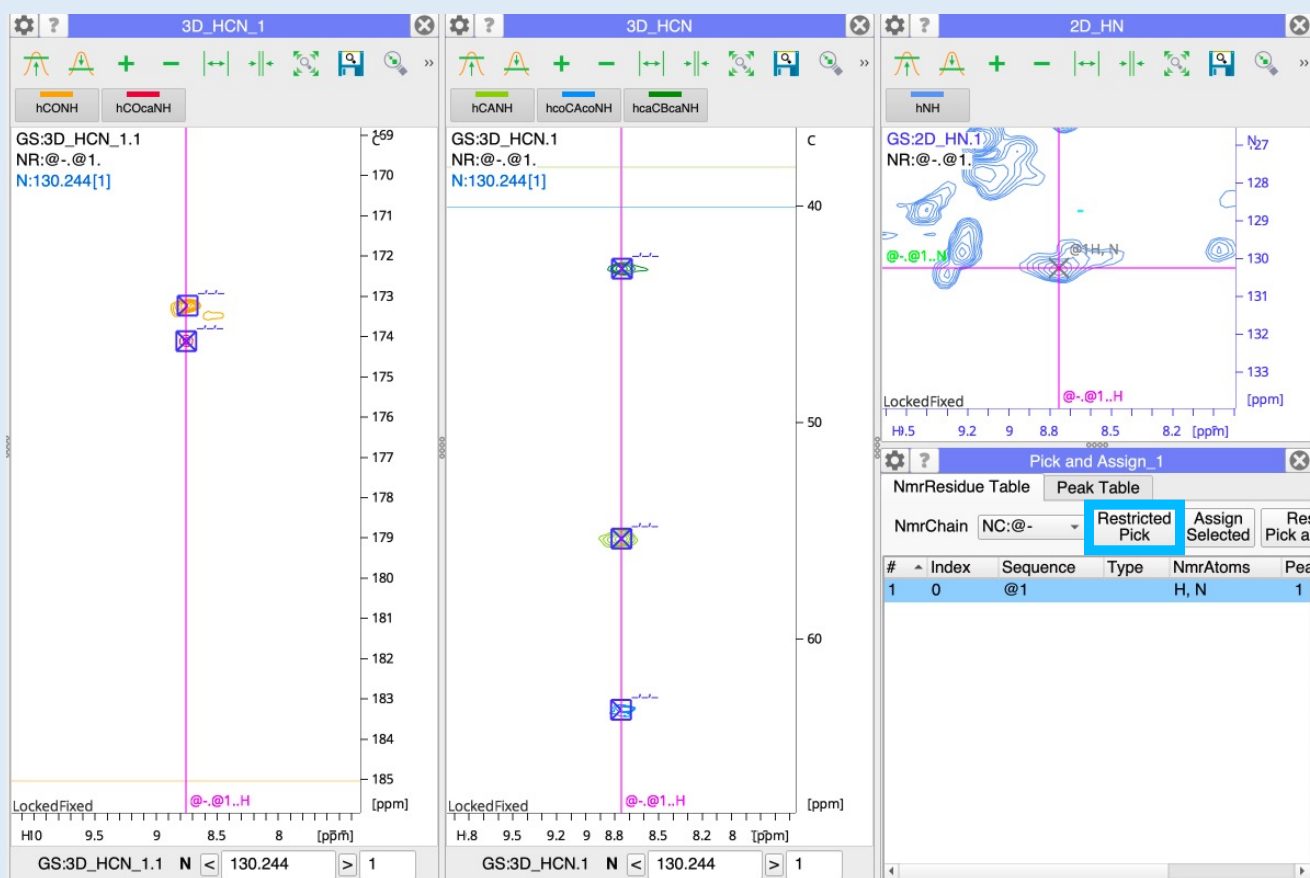
2D Navigate spectra

In the Pick and Assign module:

- **Double-click** on the row for the NmrResidue you have just created in the 2D hNH spectrum, e.g. @1.

This will cause the 2D SpectrumDisplay to focus on the position of the peak labelled @-..@1..H,N and to mark this position with two labelled rulers corresponding to each dimension. Likewise, the two 3D SpectrumDisplays will navigate to the corresponding z (i.e. N) position and mark the appropriate frequency along the H axis with a labelled ruler.

Ideally you should see peaks along this ruler in the 3D spectra.



2_E Pick 3D Peaks

If there is little overlap in the 3D spectra:

- Click the **Restricted Pick** button in the **Pick and Assign** module.

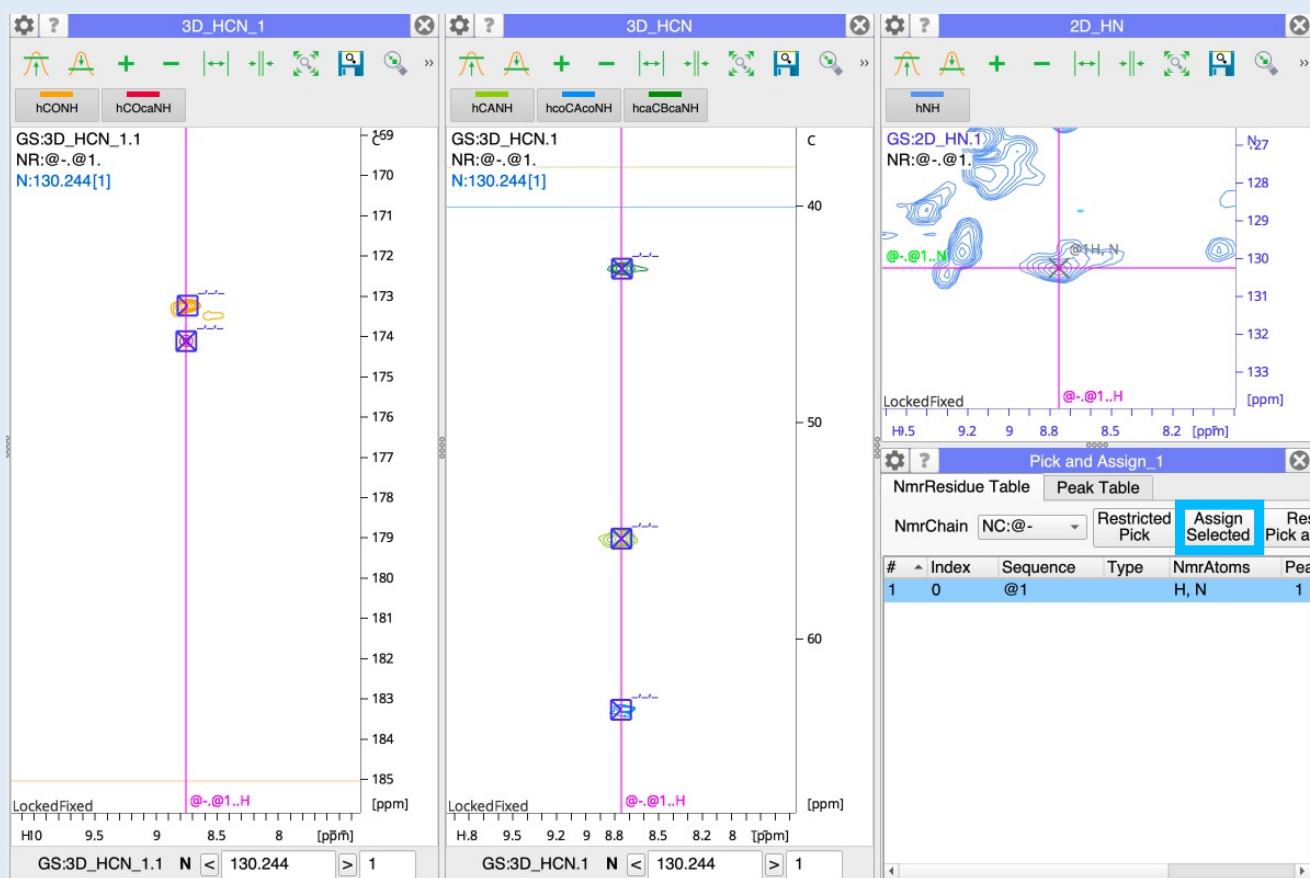
This will pick all the peaks along a narrow region in the 3D window and these peaks will be selected.

- Inspect the 3D peaks and delete any noise peaks or manually pick any peaks that weren't picked automatically (sometimes they are one plane higher or lower or slightly shifted in the H dimension).

In overlapped regions:

- Pick the peaks in the 3D spectra individually.

Be aware that the peak picking occurs on any visible peaks, so it can be helpful to adjust the contours levels to something suitable before picking your peaks.



2_F Assign 3D Peaks

Once all the peaks in the 3D spectra have been picked:

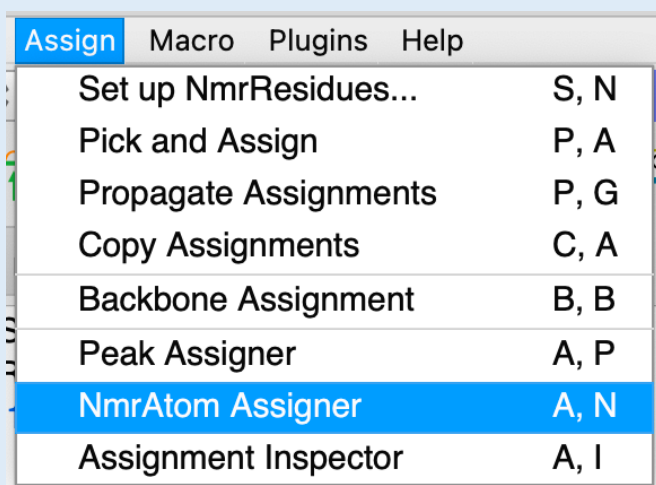
- Make sure all of them are selected. If necessary, select them by holding down **Ctrl/Cmd** and clicking the peaks or dragging the mouse over them, to select.
- Click the **Assign Selected** button in the **Pick and Assign** module.

This transfers the H and N atom assignments from the 2D spectrum to the 3D spectra and assigns the Carbon NmrAtoms.

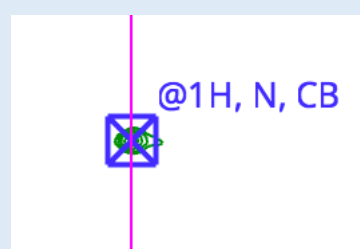
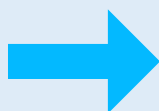
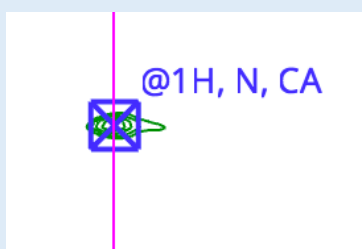
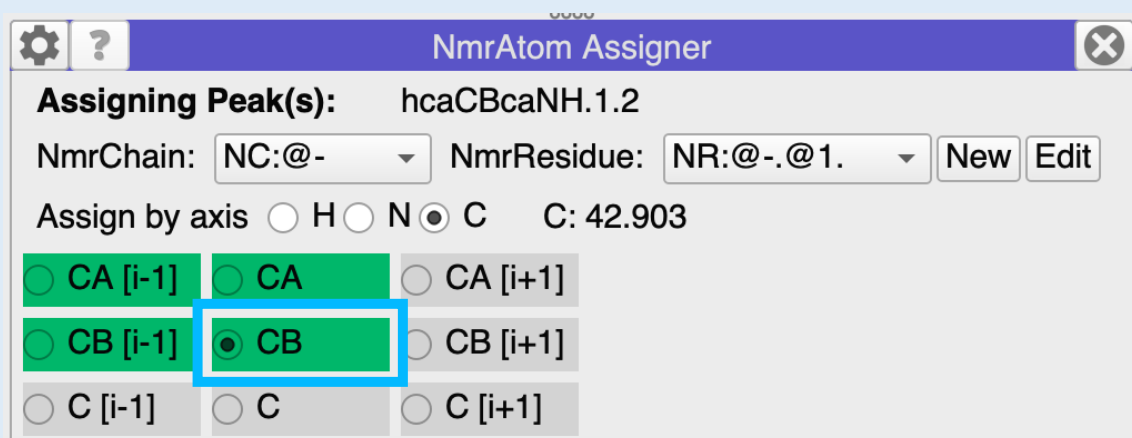
Note that some Carbon NmrAtoms are i-1 to the NH group they are linked to. This produces an additional @x-1 NmrResidue in the table:

#	Index	Sequence	Type	NmrAtoms
1	1	@1		C, CA, CB, H, N
2	0	@1-1		C, CA





or
shortcut
AN



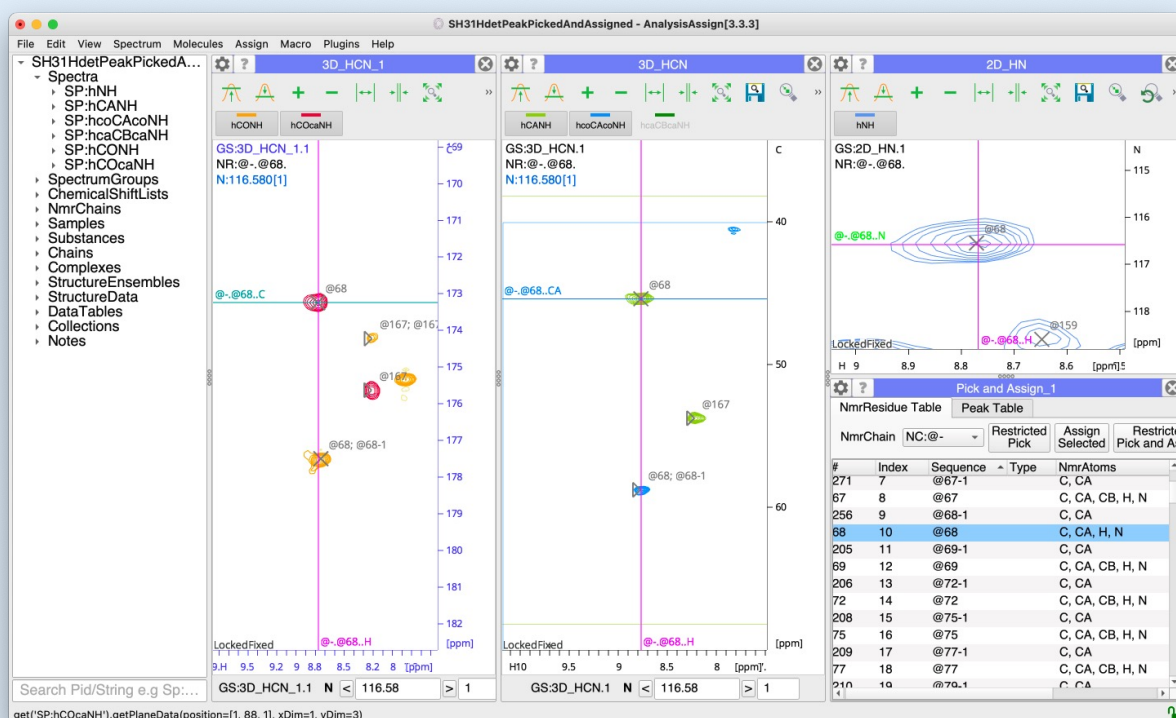
2G Correcting carbon atom type assignments

Occasionally you may find that the Carbon atom types have been assigned incorrectly (e.g. if peaks are overlapped, too many peaks have been picked or the peak intensities are unusual). In this case you can make corrections using the **NmrAtom Assigner** module:

- Go to **Main Menu** → **View** → **NmrAtom Assigner** or use the shortcut **AN**.
- Select a peak in one of the 3D windows.

The NmrAtom Assigner will predict the assignment for the carbon dimension, using green for likely and orange for less likely assignments.

- Toggle any NmrAtom button to assign/deassign it to the selected peak(s).



2H Continue to Pick and Assign the 3D spectra

To use the sequential Backbone Assignment tools in AnalysisAssign, the *i* and *i*-1 assignments for all NmrResidues need to be provided, so the procedure described in sections 2B–G above should be carried out for all peaks in the hNH spectrum, where possible.

Continue moving on to other peaks in the hNH spectrum or move through the *z*-planes of the 3D spectra to identify, pick and assign the peaks. In some cases you will find that the 3Ds do not contain peaks at the 2D peak positions – maybe because the peaks are too weak or because they derive from sidechains which don't have peaks in the 3Ds.

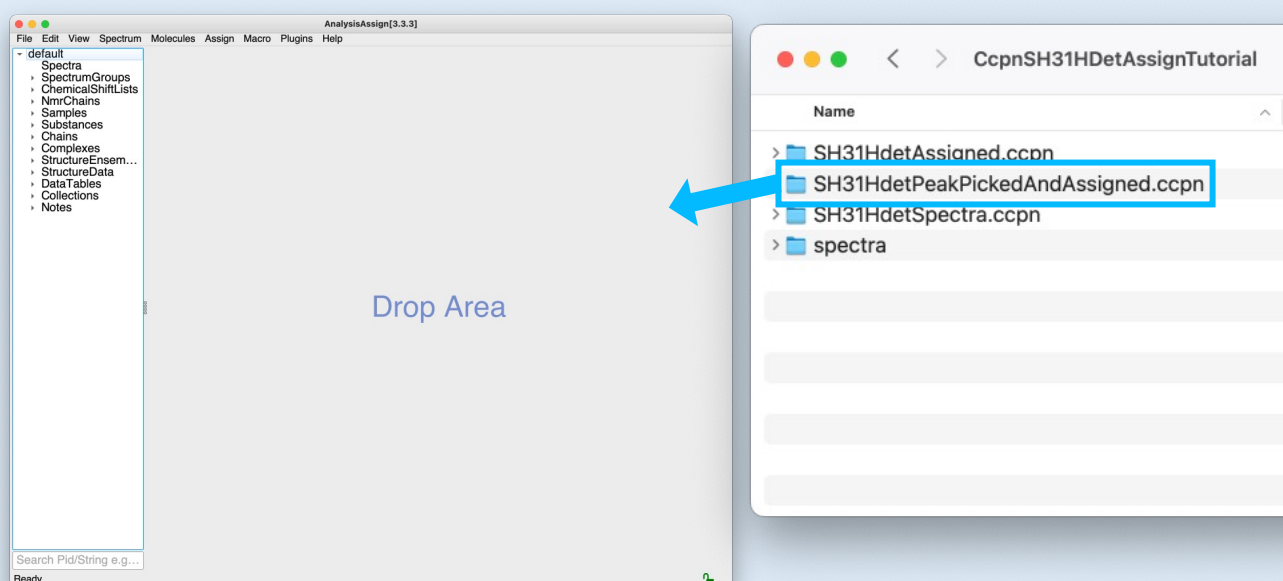
Once all three dimensions of all 3D peaks have been assigned to the appropriate NmrAtoms, the backbone assignment can be carried out, but obviously for this tutorial we provide you with a CcpNmr project in which this has already been completed.

Note that in principle, you can also Pick and Assign multiple rows (even the whole table) in one go:

- Use the **Shift** or **Ctrl/Cmd** key to select multiple rows.
- Click on **Restricted Pick** or **Restricted Pick and Assign** to pick and/or assign the peaks for all selected NmrResidues in one go.

This can work well for small proteins studied by solution NMR, but for this dataset, we suggest a more manual approach.

3 Sequential backbone assignment



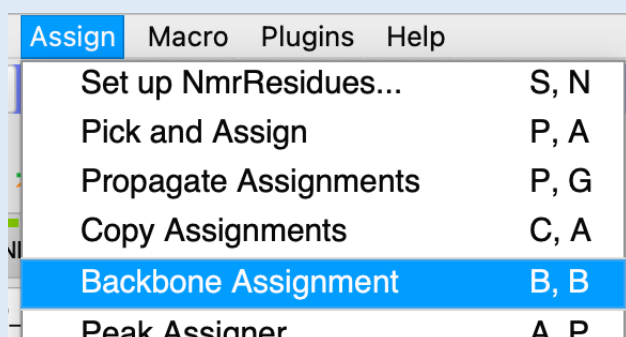
3_A Open SH31HdetPeakPickedAndAssigned.ccpn

SH31HdetPeakPickedAndAssigned.ccpn is a project in which all the carbon atom type assignments for the NmrResidues have been completed and thus can be used directly for the sequential backbone assignment. The project can be found in the CcnpnSH31HDetAssignTutorial directory.

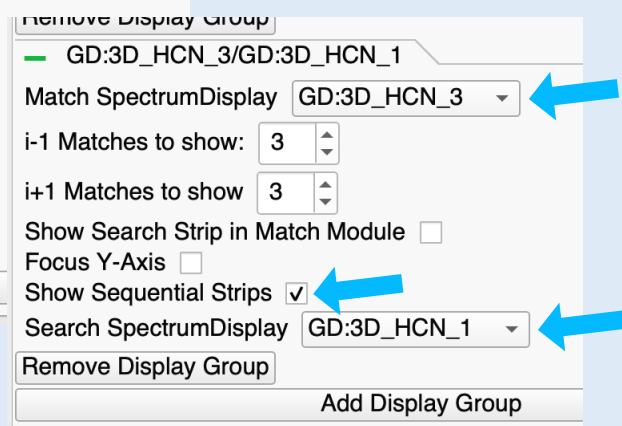
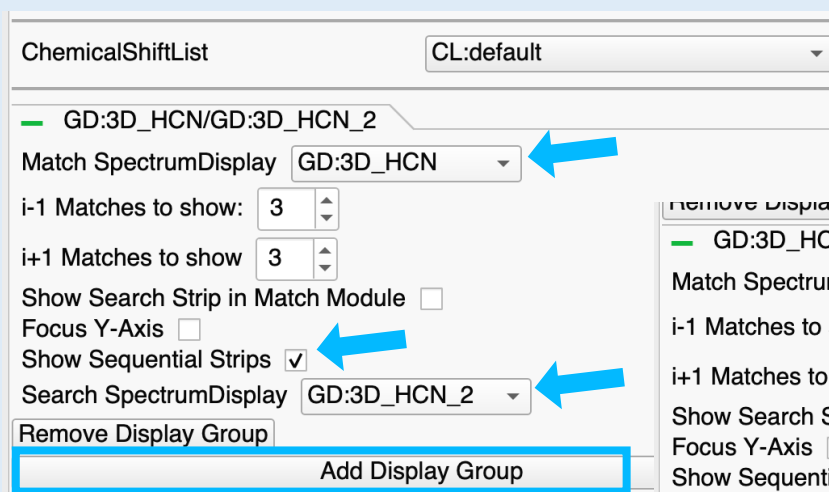
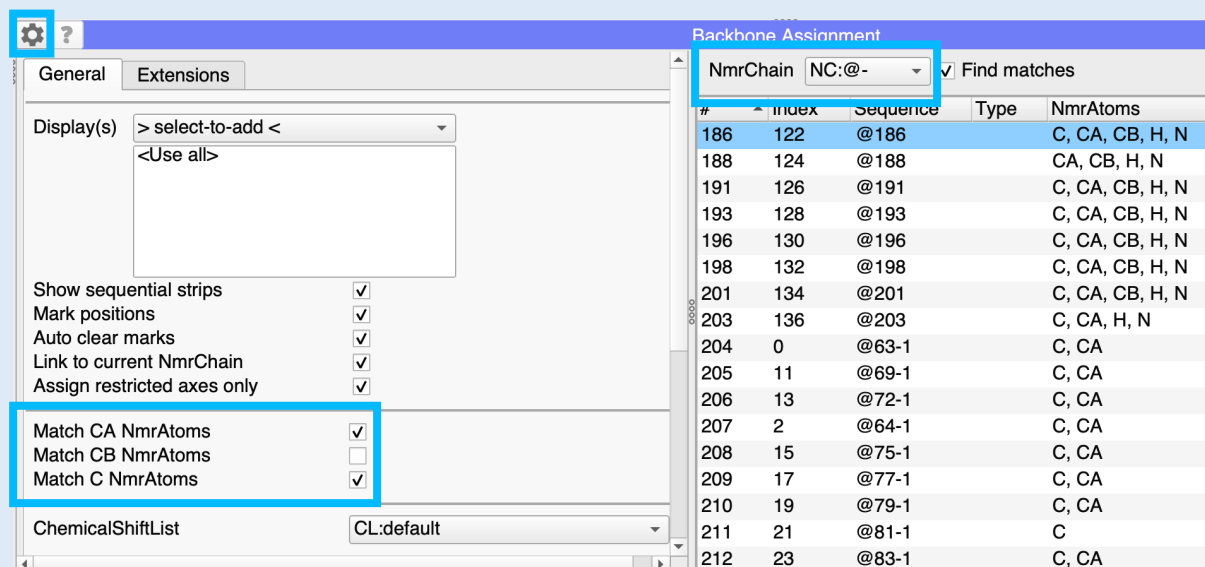
- Open the SH31HdetPeakPickedAndAssigned.ccpn project by dragging and dropping it into the Drop Area or on the sidebar. Quit the current project with or without saving.



3 Sequential backbone assignment



or shortcut
BB

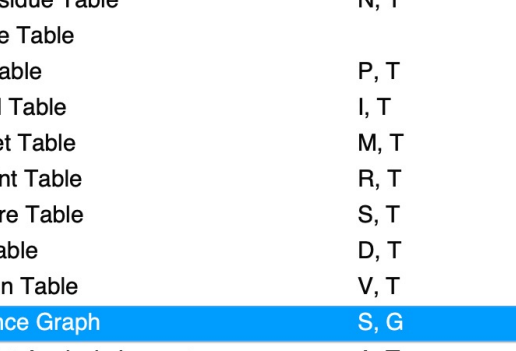


3B Open and set up the backbone assignment module

- Main Menu → Assign → Backbone Assignment or shortcut BB
- Select the NmrChain: NC:@-.
- Open the settings (gear icon) and set up as shown above, clicking on **Add Display Group** to add a second set of Match and Search SpectrumDisplays.

Note that the first display group uses the top two SpectrumDisplays, the second one uses the lower two. The right hand one is always specified as the **Match SpectrumDisplay** and the left one as the **Search SpectrumDisplay**.

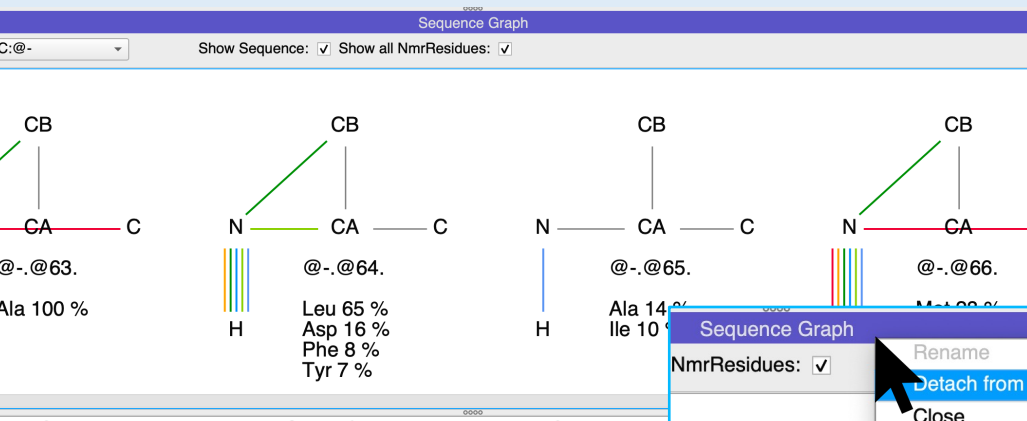
3 Sequential backbone assignment



View Spectrum Molecules Assign Macro Plugins Help

Chemical Shift Table	C, T
NmrResidue Table	N, T
Residue Table	
Peak Table	P, T
Integral Table	I, T
Multiplet Table	M, T
Restraint Table	R, T
Structure Table	S, T
Data Table	D, T
Violation Table	V, T
Sequence Graph	S, G
Restraint Analysis Inspector	A, T

or Shortcut SG



Sequence Graph

NmrChain: NC:@- Show Sequence: ☒ Show all NmrResidues: ☒

CB

N — CA — C

@-.@63.

Ala 100 %

H

CB

N — CA — C

@-.@64.

Leu 65 %
Asp 16 %
Phe 8 %
Tyr 7 %

H

CB

N — CA — C

@-.@65.

Ala 14 %
Ile 10 %

H

CB

N — CA — C

@-.@66.

Met 99 %

H

SH3:A MDETGKELVL₁₀ ALYDYQEKSP₂₀ REV TMKKGDI₃₀ LT

Sequence Graph

NmrResidues: ☒

Rename
Detach from Drop Area
Close
Close Others
Close All
Copy Gid to clipboard

3c Open and set the Sequence Graph module

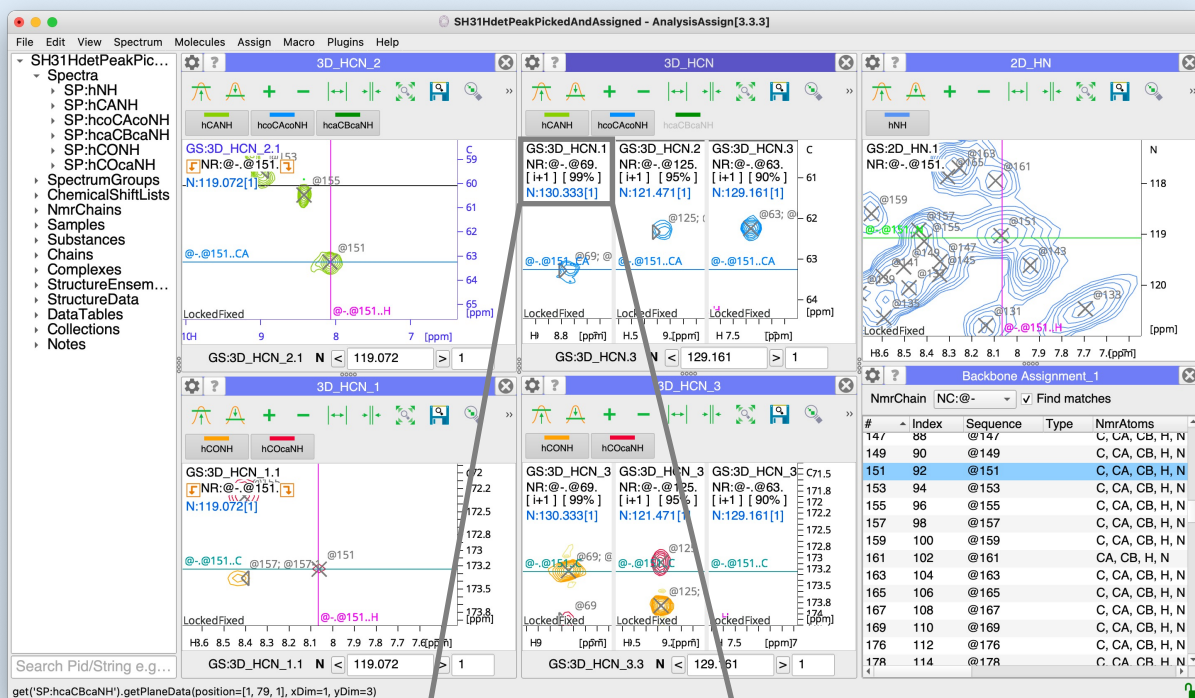
- **Main Menu → View → Sequence Graph** or shortcut **SG**
- Select the NmrChain: NC:@-
- Rearrange the modules to a layout like the Figure below
(or detach the Sequence Graph module and keep separate).



3 Sequential backbone assignment

#	Index	Sequence	Type	NmrAtoms	Peak count
66	6	@66		C, CA, CB, H, N	6
67	8	@67		C, CA, CB, H, N	6
68	10	@68		C, CA, H, N	6
69	12	@69		C, CA, CB, H, N	6

Double-click on row @151



NmrAtom

Strip ID

match direction

match probability

¹⁵N chemical shift

3D Assign in the i+1 direction ...

- Double-click the @151 row.

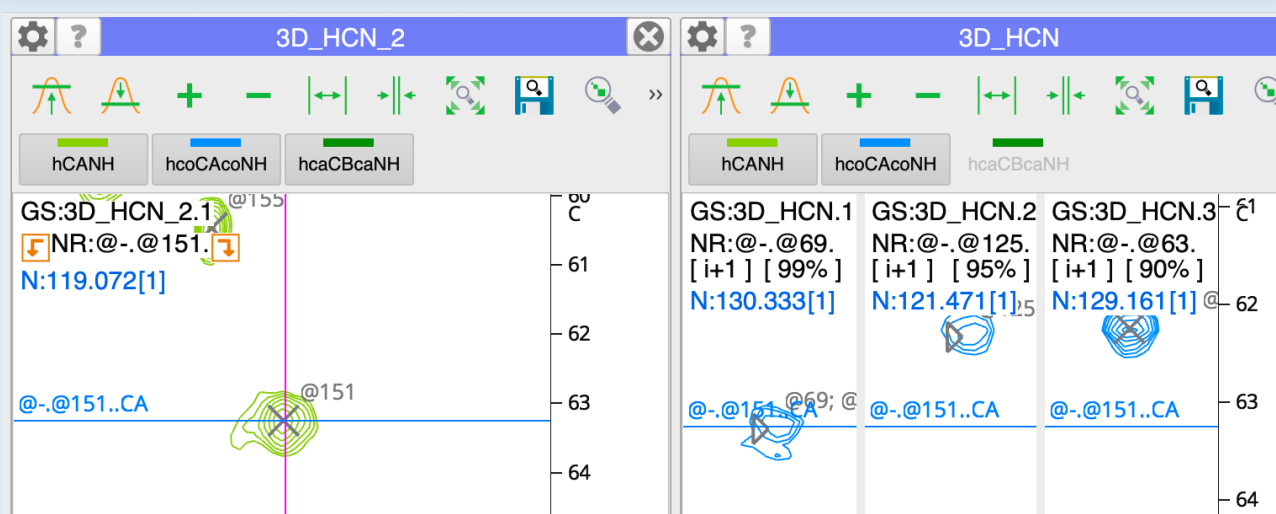
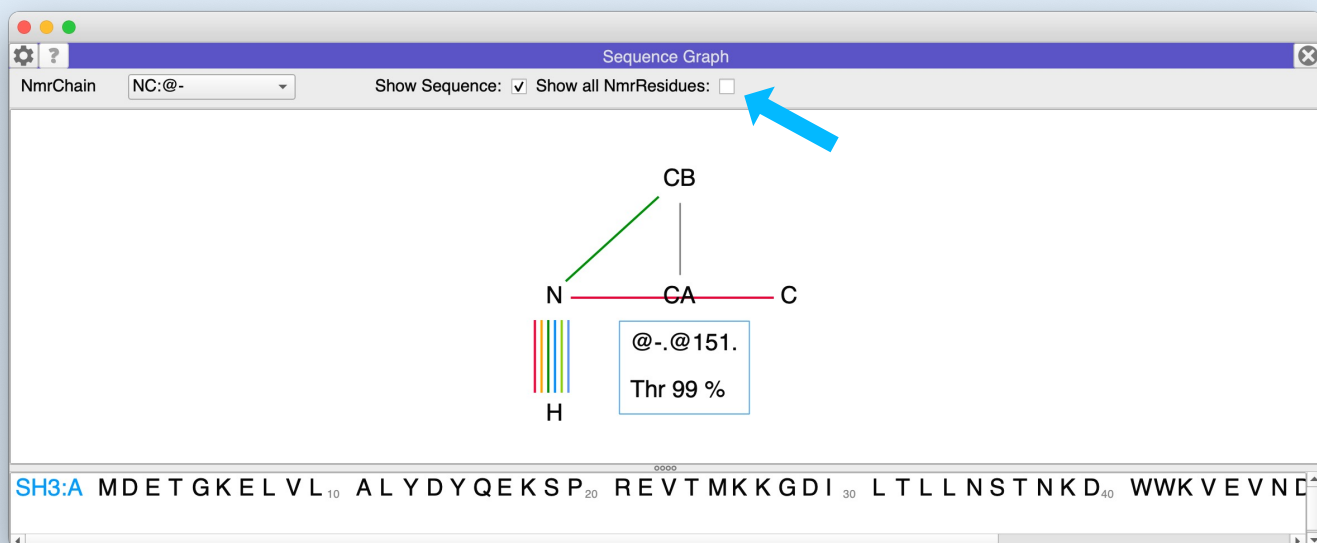
You will see a series of changes in the GUI:

2D_HN: navigates to the @151 peak in the hNH spectrum (the H and N chemical shifts are marked)

Search Displays (3D_HCN_2, 3D_HCN_1): navigate to the @151 NH plane and mark the CA and C chemical shifts.

Match Displays (3D_HCN, 3D_HCN_3): display 3 strips in the order (left to right) that the AnalysisAssign algorithm thinks best match the @151 CA and C chemical shifts. At the top of each strip you will see the StripID, the NmrAtom associated with that strip, the link direction and match probability and the Nitrogen chemical shift in the z-plane.

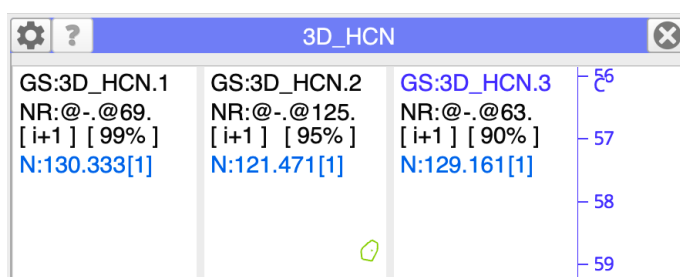
3 Sequential backbone assignment



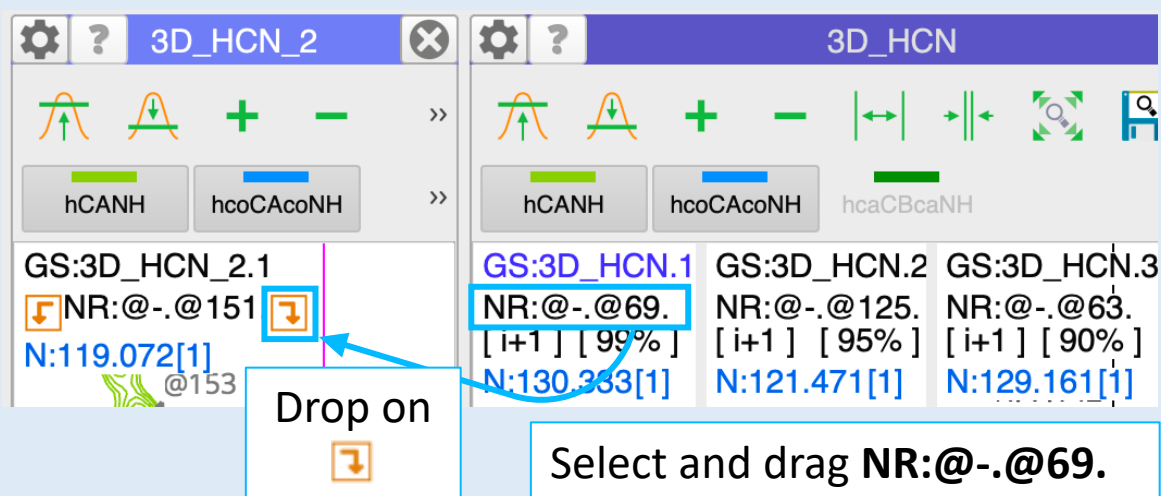
3D ... Assign in the i+1 direction

- If you deselect **Show all NmrResidues** in the SequenceGraph:
Then **Sequence Graph** will draw a schematic residue, labelled with the NmrResidue name (@151) and predictions of the possible residue type(s) below it. You can see @151 is predicted to be a Threonine.
The CA chemical shift of a Threonine is at around 58–68 ppm, so you may find that you need to shift the y-axis a little lower in the top two modules to see the CA search and match peaks.

If you are doing this on a small screen (e.g. on a laptop), then it can be helpful to remove the toolbars in the SpectrumDisplays by typing **TB** and **SB** in each SpectrumDisplay.



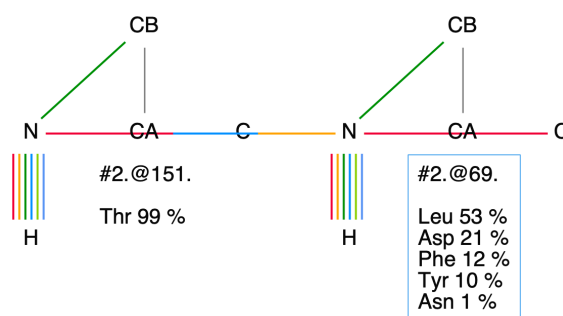
3 Sequential backbone assignment



3E Connecting NmrResidues

Upon examination, the first strip in the Match SpectrumDisplays (3D_HCN, 3D_HCN_3) shows NmrResidue @69; the H, CA and C marks align with the peaks of this residue, we can therefore see that is the best match for @151.

- Select the strip label **NR:@-.@69.** on a Match SpectrumDisplay
- Drag and drop the label on to the  icon in a Search SpectrumDisplay
- Check **Show all NmrResidues** in the **Sequence Graph** module (and make sure NmrChain #2 is selected) to see the new connection



At this point @151 and @69 have been put into a so-called connected stretch and they are in a new NmrChain called #2, which means that the NmrResidue names have changed to #2.@151. and #2.@69. The Backbone Assignment module should automatically be set to NmrChain #2.

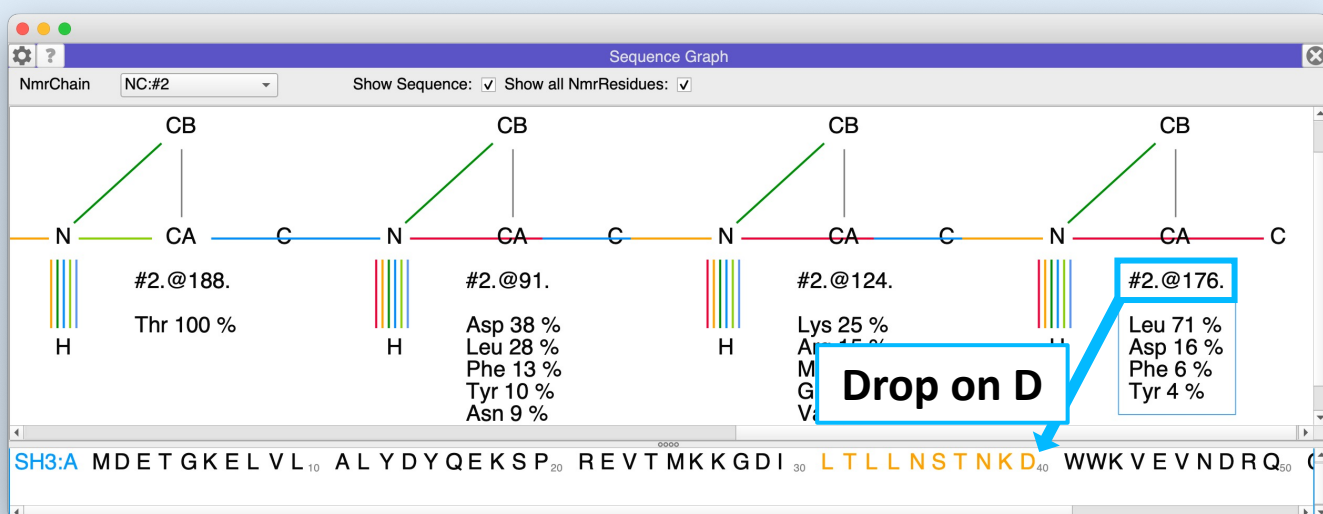
When strips are dragged and dropped through this procedure, AnalysisAssign will subsequently look for i+1 matches for @69. The algorithm thinks that @87 is a good match for @69 and on inspection it is a match, so selecting and dragging @87 onto the  of a Search Spectrum Display will continue the assignment.

Note: The Settings panels of both the **Sequence Graph** and **Backbone Assignment** modules contain a setting **Link to current NmrChain**. If this is checked, then selecting a new NmrChain in one module will automatically update that used in the other module. You can switch this on or off as suits you best.

Link to current NmrChain



3 Sequential backbone assignment



3F Link NmrChain to Chain

If you continue assigning in the i+1 direction, you should end up with a stretch consisting of:

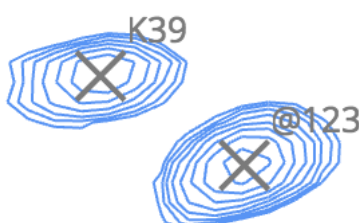
@151 @69 @87 @178 @95 @188 @91 @124 @176

and **L T L L N S T N K D** will be highlighted in the Sequence Graph sequence because this is the best match in the sequence based on the chemical shifts.

- Select e.g. the label **#2.@176** in the Sequence Graph and drag it onto the right-hand residue (**D**) in the highlighted sequence
- Click **Yes** in the confirmation popup

You may notice a new NmrChain **A** has been automatically created and is linked to the Chain **A** for the above NmrResidues–Residues.

Now the relevant hNH peaks for the selected stretch are fully assigned. You can inspect this in the **2D_HN** Spectrum Display and see some fully assigned peaks, e.g. K39.



3 Sequential backbone assignment

The screenshot shows the NMR assignment software interface. At the top, the 'Backbone Assignment_1' window displays a table of NmrResidues. A blue arrow points to the 'Index' column header, and a text box says 'Double-click on row 32-1'. Below this, two '3D_HCN' windows show match spectrum displays. A blue arrow points to the 'NR:@-.@72.' match in the '3D_HCN.1' window, and a text box says 'Drop on' with a drop icon. At the bottom, the 'Sequence Graph' window shows a protein sequence with assigned residues highlighted in blue: A.21.ARG, A.22.GLU, A.23.VAL, and A.24.THR. The sequence is: SH3:A MDET GKELVL₁₀ ALYDYQEKSP₂₀ REVTMKKGDI₃₀ LTLNSTNKD₄₀ WWKVEVNDRO₅₀ GF.

3G Assign in the i-1 direction

The Sequential backbone assignment in the i-1 direction is fully analogous to the i+1 direction except you will need to start with a '-1' NmrResidue and drop the strip label on the  icon.

- Go to the Backbone Assignment module table
- Click on the **Index** column header to sort the NmrResidues by index.
- Double-click on row 28-1
- Select the best match **NR:@-.@72.** in one of the Match SpectrumDisplays (3D_HCN or 3D_HCN_3) and drag & drop it onto the  icon in a Search SpectrumDisplay (3D_HCN_1 or 3D_HCN_2).

Note that as this will automatically assign NmrResidue @72 to 31.LEU.

- Continue in this direction until you have added @137 @123, @68, @121, @97, @125, @169, @193, @112 and have assigned everything up until Proline 20.

Note that you cannot continue the assignment beyond this point, since Prolines do not have an NH group and thus always result in a break in the sequential link of residues via NH based spectra.

If you wish, continue trying to assign other stretches of the protein.

Peak Assigner

Current Peak: hncacb.1.76

H: 8.597 C: 52.496 N: 124.006

NmrAtom	Delta
A.25.ARG.H	0.001

Alternatives

NmrAtom	Delta
A.74.VAL.H	0.003

Alternatives

NmrAtom	Delta
A.26-1..CA	0.013
A.25.ARG.CA	0.039
A.50.ALA.CA	0.087
A.51-1..CA	0.089
A.65.ALA.CA	0.368
A.66-1..CA	0.379
A.67.ASN.CA	0.382
A.68-1..CA	0.390

Alternatives

NmrAtom	Delta
A.51.GLU.N	0.017
A.42.CYS.N	0.142
A.50.ALA.N	0.184

Currently selected peak(s)

Peak dimensions with isotopes and chemical shifts

Double-click to de-assign

Double-click to assign

The **Peak Assigner** is very versatile and it has a number of options to guide its behaviour. Use the gear box icon to show/hide these. Crucial parameters are the assignment tolerances along each dimension for the different spectra. These are set to (sensible) default values but can also be modified by using the **Dimensions** tab of the **Spectrum Properties** popups of the respective spectra (open by double-clicking on a spectrum in the sidebar).

4A Editing Assignments

An easy way to edit the assignment of a peak is with the Peak Assigner:

- Bring up the **Peak Assigner** module either by going to **Main Menu** → **View** → **Peak Assigner**, with shortcut **AP**.

The upper panels show the peak assignments, the lower ones show NmrAtoms with Chemical Shifts close the peak position. There is a column for each peak dimension. The Isotope type and chemical shift are shown.

- **Double-click** on an NmrAtom in the top panel to remove this assignment from that dimension of the selected peak.
- **Double-click** on an NmrAtom in the lower middle panel to assign this NmrAtom to the selected peak in that dimension.
- Use the **New** button to add a new NmrAtom not in the list, or **Edit** to edit the selected NmrAtom.

Assign	Macro	Plugins	Help
Set up NmrResidues...	S, N		
Pick and Assign	P, A		
Propagate Assignments	P, G		
Copy Assignments	C, A		
Backbone Assignment	B, B		
Peak Assigner	A, P		
NmrAtom Assigner	A, N		
Assignment Inspector	A, I		

or
shortcut
AI

Assignment Inspector						
ChemicalShiftList CL:default						
Value (ppm)	Value Error (ppm)	NmrAtom	ResidueType	AtomName	Peak Count	Total Peak Count
27.927	None	NA:A.19.IHP.CB	IHP	CB	1	1
29.667	None	NA:A.64.GLN.CB	GLN	CB	1	
59.188	3.346	NA:A.25-1..CA	None	CA	5	
66.475	1.780	NA:A.21-1..CA	None	CA	5	

Single click

Filter by NmrAtom(s): A.25-1..CA							
#	Assign F1	Assign F2	Assign F3	Pos F1	Pos F2	Pos F3	LW F1
76	A.25.ARG.H	A.25-1..CA	A.25.ARG.N	8.597	52.496	124.006	20.823
75	A.25.ARG.H	A.25-1..CA	A.25.ARG.N	8.598	60.852	123.986	22.097
42	A.25.ARG.H	A.25-1..CA	A.25.ARG.N	8.597	60.860	123.999	21.756
22	A.25.ARG.H	A.25-1..CA	A.25.ARG.N	8.597	60.864	123.999	20.901
37	A.25.ARG.H	A.25-1..CA	A.25.ARG.N	8.596	60.868	123.993	21.719

4B Checking Assignments

The Assignment Inspector is a useful tool for inspecting assignments:

- Open the Assignment Inspector module:

Main Menu → Assign → Assignment Inspector or shortcut **AI**

- Make sure your ChemicalShiftList of interest is selected.
- Click** on **Value Error** to sort the top table by the Chemical Shift error.
Chemical Shifts with unusually high errors should be investigated.
- Select a row in the (top) Chemical Shift table.
This will populate the peak table below with all peaks belonging to that NmrResidue (not NmrAtom!).
- You can filter the peak table by clicking on an NmrAtom on the left.
- Clicking** on **Pos F2** will sort the peak table by the carbon chemical shift and you can find outliers.

Double click

The screenshot shows the 'Assignment Inspector' window. On the left, there are tabs for 'General', 'NH Strips', and 'CH Strips'. The 'NH Strips' tab is active, showing a list of display groups: 'GD:3D_HCN' and 'GD:3D_HCN_1'. Below this, there are controls for 'i-1 strips to show' and 'i+1 strips to show', both set to 1. The main area displays a table of chemical shifts. A blue box highlights the left sidebar, and a blue arrow points from the 'Double click' text to a row in the table.

ChemicalShift	Value (ppm)	Value Error (ppm)	NmrAtom	ResidueType	AtomName
CS:default.383	56.251	None	NA:A.23-1..CA	None	CA
CS:default.331	171.602	None	NA:A.23.VAL.C	VAL	C
CS:default.201	59.758	None	NA:A.23.VAL.CA	VAL	CA
CS:default.447	34.967	None	NA:A.23.VAL.CB	VAL	CB
CS:default.133	7.398	0.010	NA:A.23.VAL.H	VAL	H
CS:default.132	112.272	0.106	NA:A.23.VAL.N	VAL	N
CS:default.258	171.525	None	NA:A.24-1..C	None	C

Below the table, there is a section titled 'Filter by NmrAtom(s): Peaks assigned to NmrAtom(s): <Use all>'. It lists several NmrAtom entries: A.23.VAL.C, A.23.VAL.CA, A.23.VAL.CB, A.23.VAL.H, and A.23.VAL.N. To the right of these entries is a table of peaks assigned to them.

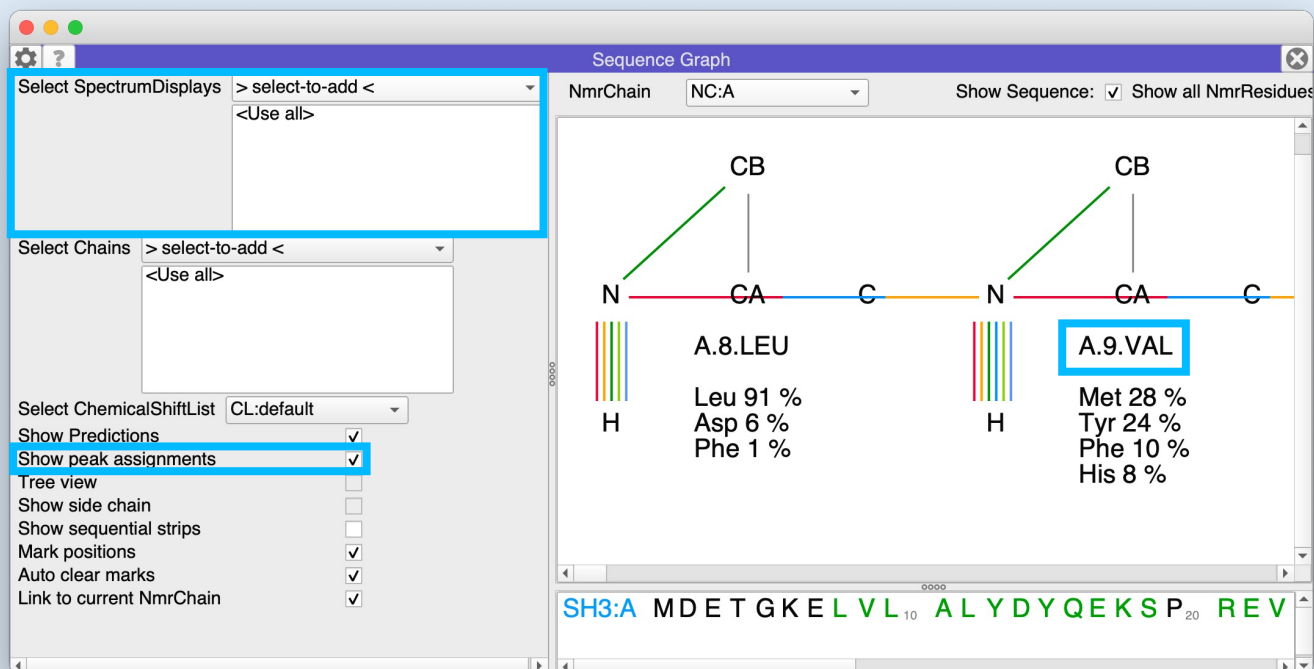
#	Spectrum	Assign F1	Assign F2	Assign F3	Pc
51	hcoCAcoNH	A.23.VAL.H	A.23.VAL.N	A.23-1..CA	7.
63	hCOcaNH	A.23.VAL.H	A.23.VAL.N	A.23.VAL.C	7.
67	hcaCBcaNH	A.23.VAL.H	A.23.VAL.N	A.23.VAL.CB	7.
69	hNH	A.23.VAL.H	A.23.VAL.N	None	7.

4C Navigating from the Assignment Inspector

The Assignment Inspector can be used for easy navigation so you can inspect the spectra for certain regions of the protein quickly:

- Open the Assignment Inspector Settings
- Go to the **NH Strips** tab
- Select the Displays to which you wish to navigate
- Select how many i-1/i+1 strips you wish to see
- If required, click on **Add Display Group** to add a second Display Group in which you can select different parameters.
- **Double-click** on an NmrResidue block in the Chemical Shift (top) table to navigate to the SpectrumDisplays selected.

Sequence Graph



SG

4D Sequence Graph

As well as being able to use the Sequence Graph to make sequence specific assignments, it also includes other information:

- Coloured lines show links between NmrAtoms from peak assignments. The colours of the lines reflect the contour colour of the spectra in which the peaks are found.

You can switch this feature off in the settings (uncheck **Show peak assignments**).

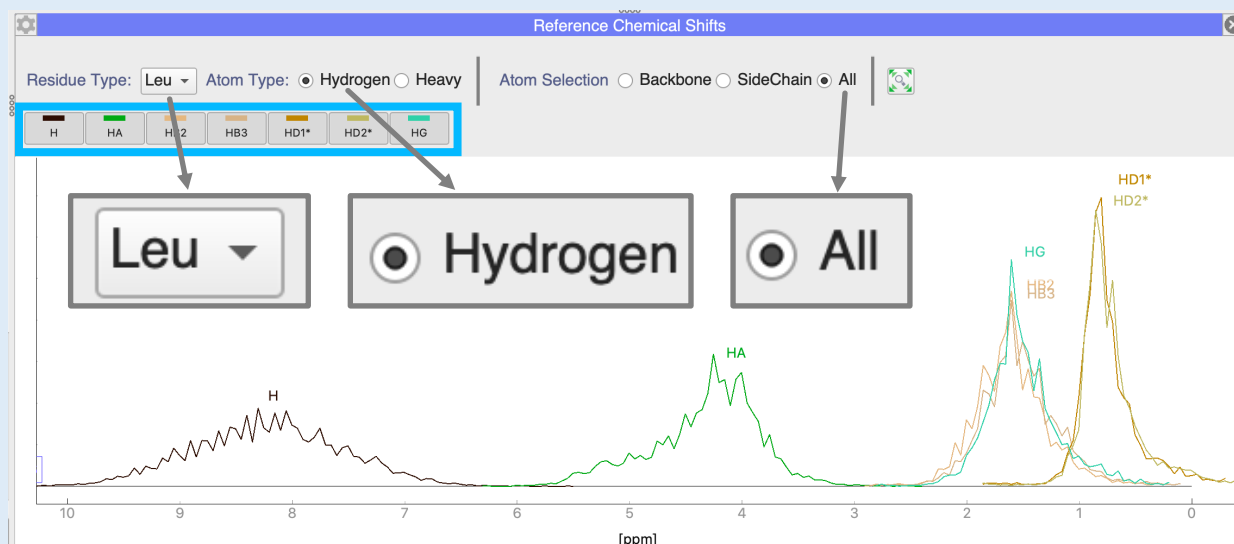
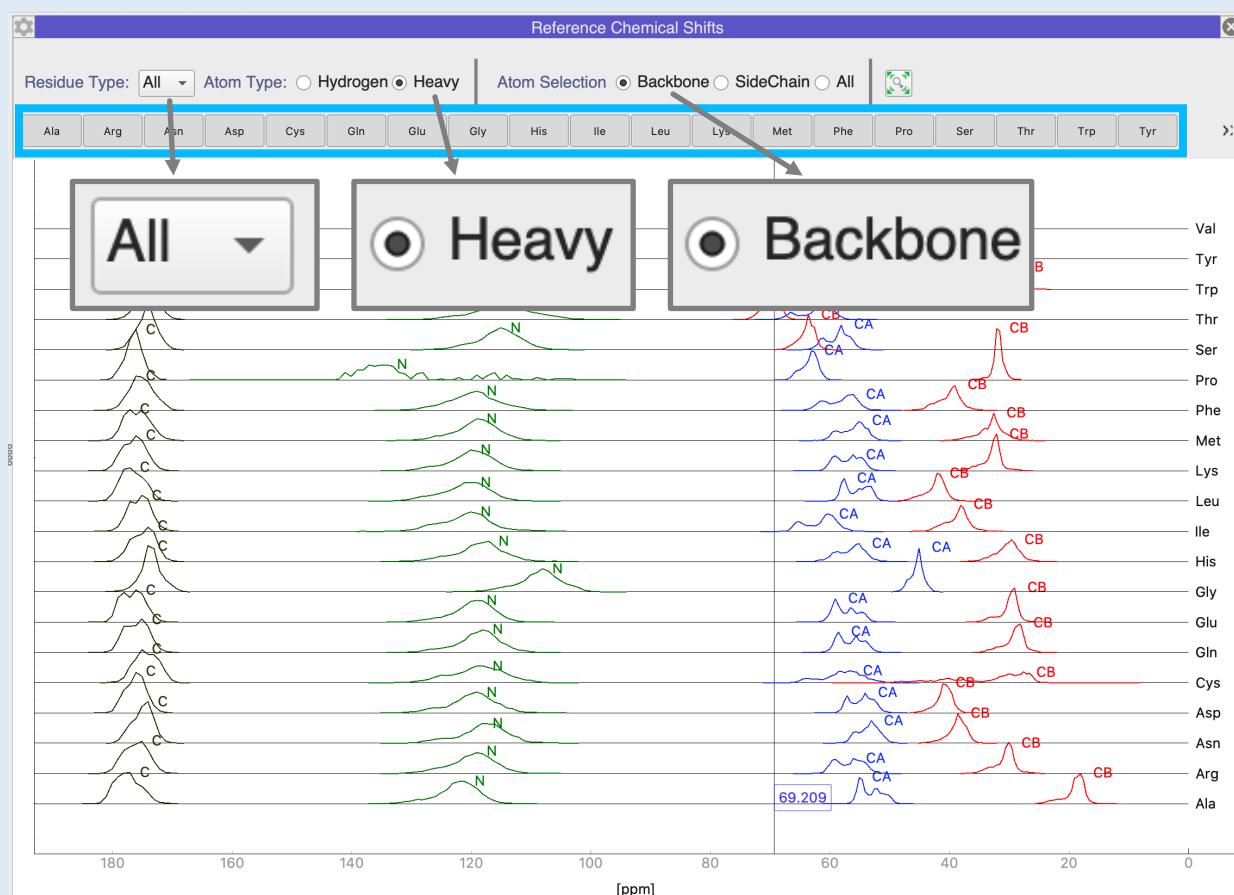
- Below each NmrResidue you will see predictions for the amino acid type. These are based on the chemical shifts and atom types of the NmrAtoms in the NmrResidues. The more information there is, the more accurate the prediction will be.

You can switch this feature off in the settings (uncheck **Show Predictions**).

- In the Settings panel you can choose (Spectrum) Displays. If at least one Spectrum Display is selected, then **double-clicking** on an NmrResidue in the Sequence Graph will place marks for that NmrResidue and navigate to its positions in the selected Spectrum Displays.

- **Right-clicking** on an NmrResidue will allow you to disconnect or edit it:

◀	Disconnect Previous NmrResidue
—	Disconnect NmrResidue
▶	Disconnect Next NmrResidue
	Disconnect all NmrResidues
	Deassign Connected NmrResidues
↶	Move NmrResidue to Front
↷	Move NmrResidue to End
	Edit NmrResidue A.12.LEU
	Show NmrResidue



RC

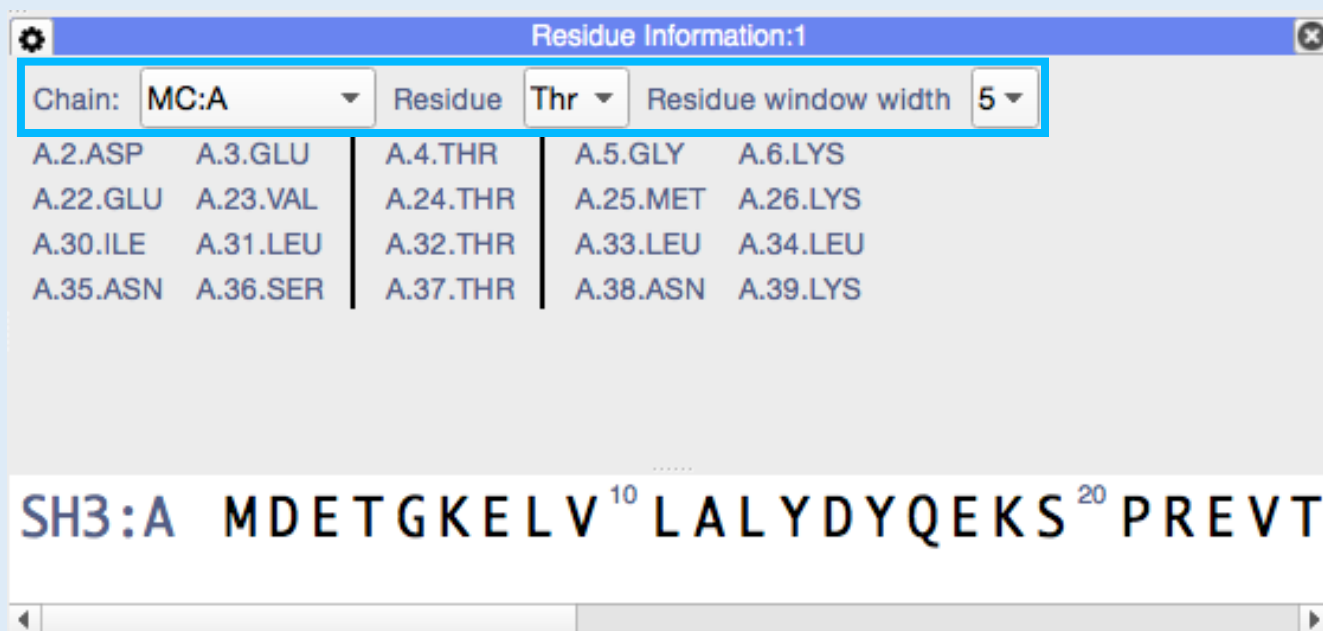
4E Reference Chemical Shifts

You can check the standard chemical shifts for protein amino acids within CcpNmr Analysis:

- Go to **Main Menu** → **Molecules** → **Reference Chemical Shifts**, or type **RC**.
- For the **Residue Type** select either **All** or an individual amino acid, e.g. **Leu**.
- For the **Atom Type** select either **Hydrogen** or **Heavy**.
- For the **Atom Selection** select **Backbone**, **SideChain** or **All**.
- Switch off particular amino acid or atom types in the toolbar.

A mouse cursor correlates the ppm position with that in your SpectrumDisplays.

You can move the graph or zoom with the mouse wheel on axes or in the main graph area like in a SpectrumDisplay.

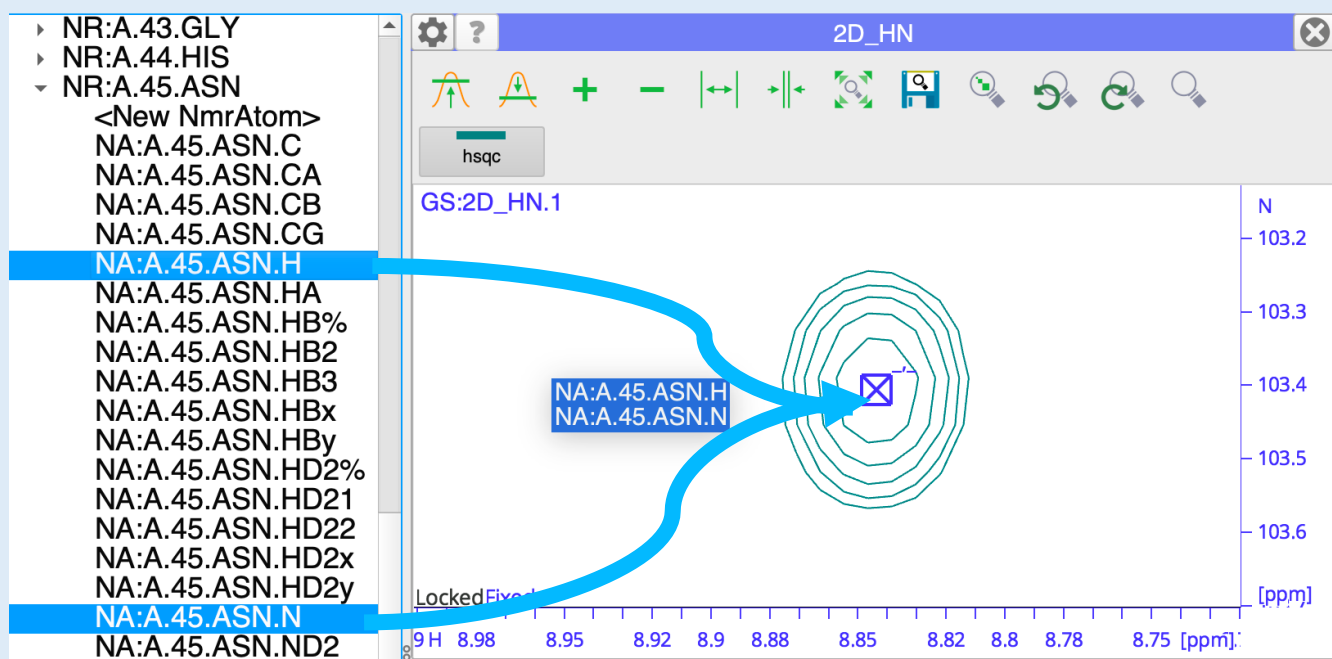


RI

4F Residue Information

You can look at different residue types in your sequence and the motifs they are contained in:

- Go to **Main Menu** → **Molecules** → **Residue Information**, or type **RI**.
- Select the **Chain**, **Residue Type** and **Residue Window Width** of your choice. The full sequence is shown below and if you have made any sequence specific assignments, then these residues will be highlighted in green.



4C Dropping NmrAtoms or NmrResidue onto a selected peak

If you have a peak for which you know the assignments, an alternative to using the Peak Assigner is to drag NmrAtoms from the sidebar onto peaks.

- In the sidebar expand the NmrChain of your choice and then the NmrResidue you are interested in.
- **Drag** one or more NmrAtoms onto the selected peak.

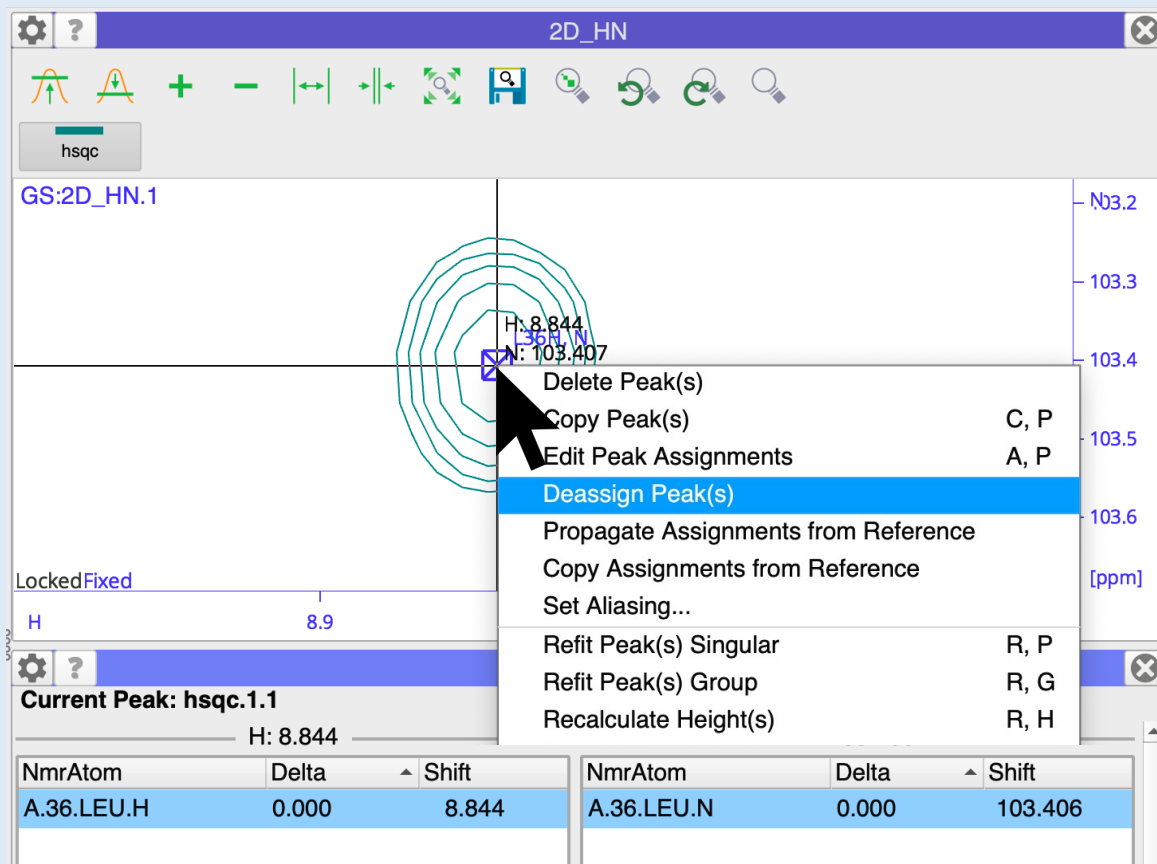
You can also drag an NmrResidue and then select which atoms to assign to the peak:

NmrAtoms assignment options

Peak(s): hsqc.1.1
NmrResidue(s): A.45.ASN

AxisCode	NmrAtoms
H	☒ H ☐ HA ☐ HB% ☐ HB2 ☐ HB3 ☐ HBx ☐ HBy
N	☒ N ☐ ND2

Close Add



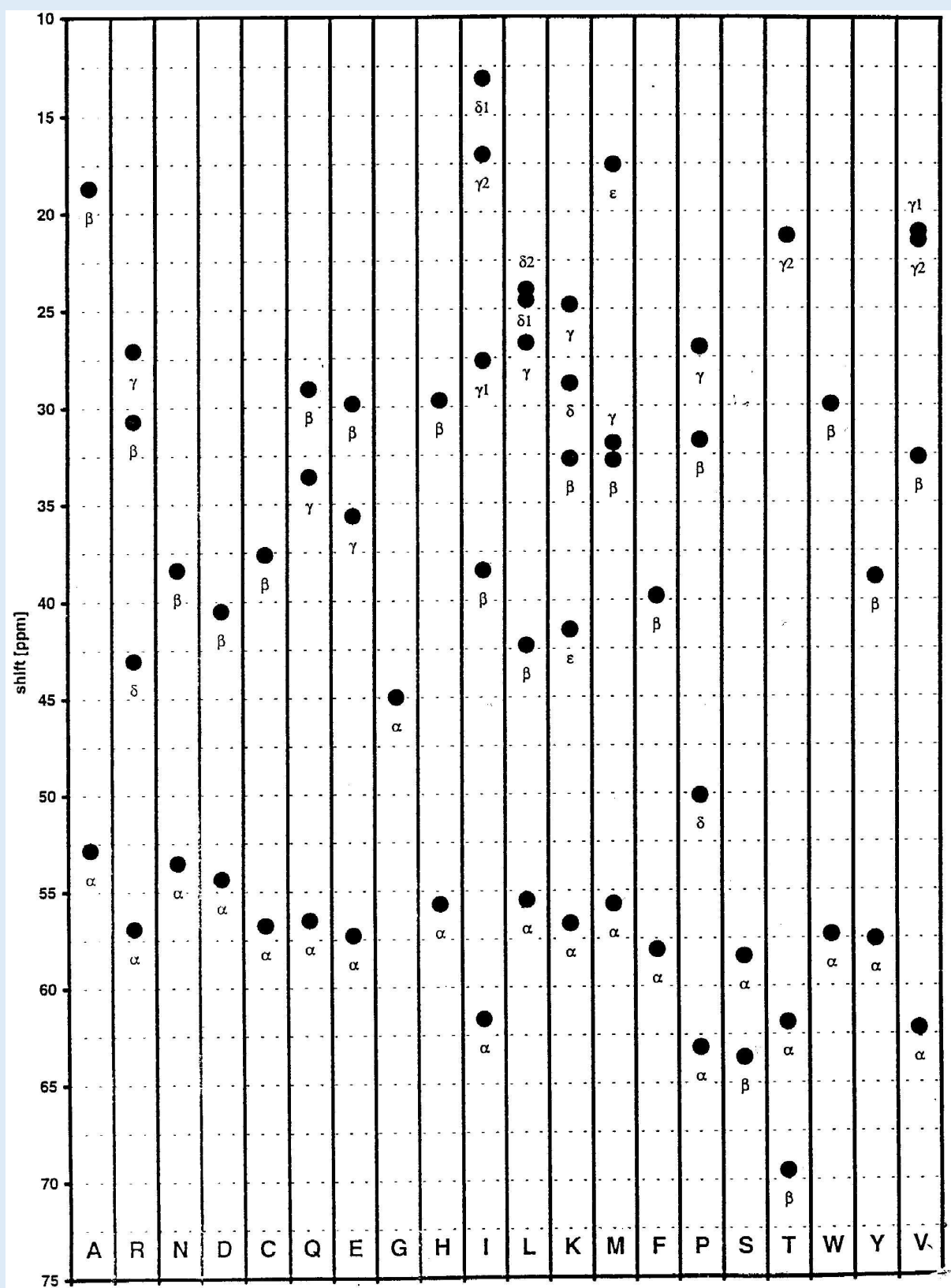
4_H Removing Assignments quickly

To remove the assignments of one or more peaks quickly:

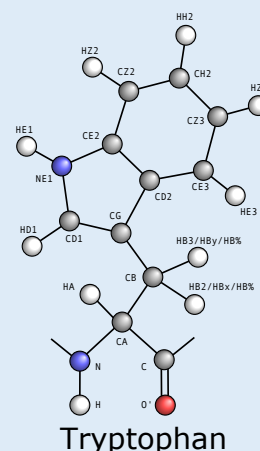
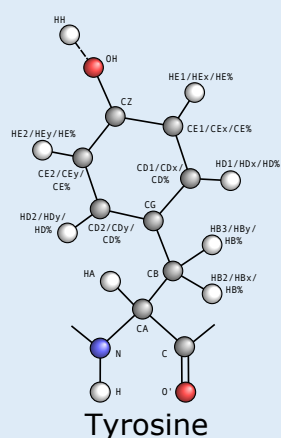
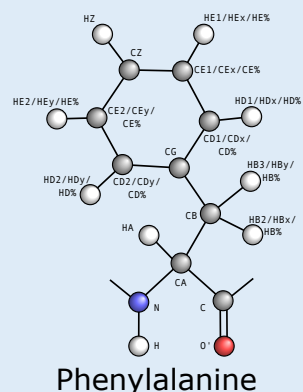
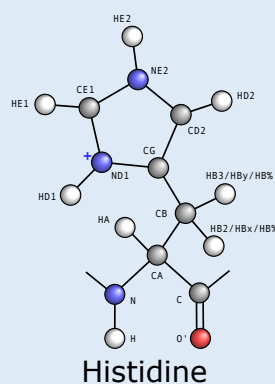
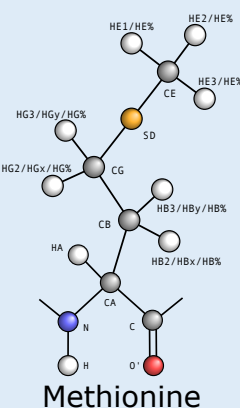
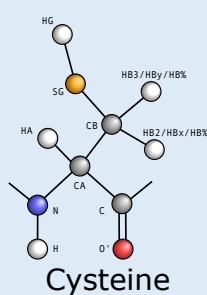
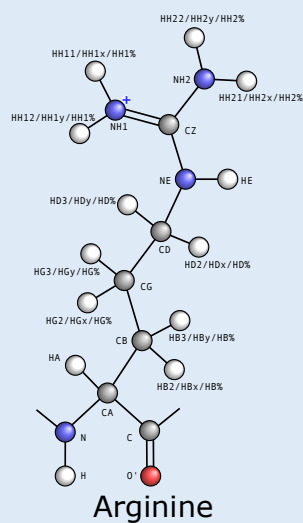
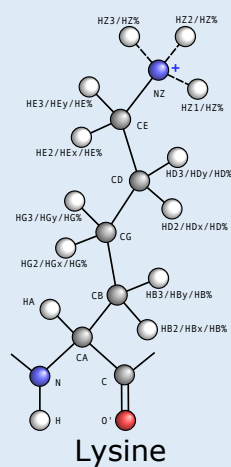
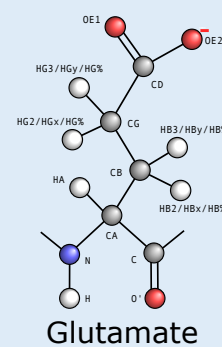
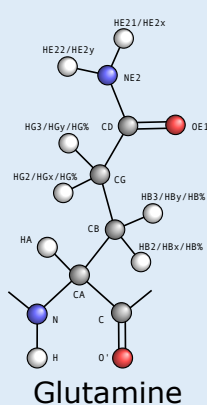
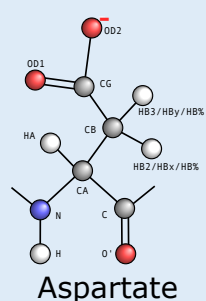
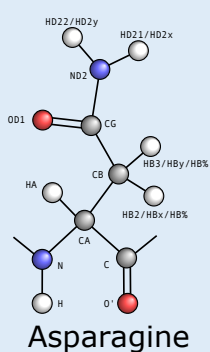
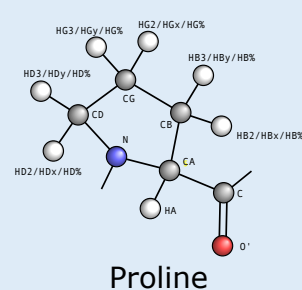
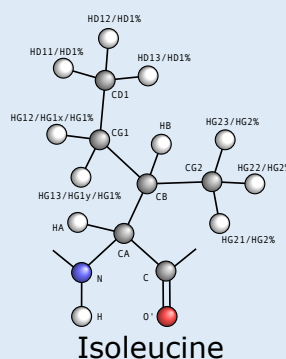
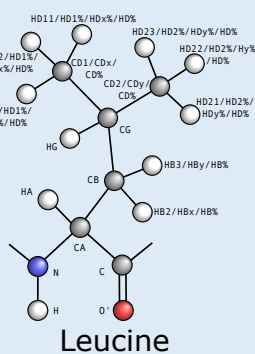
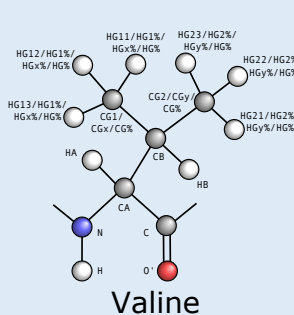
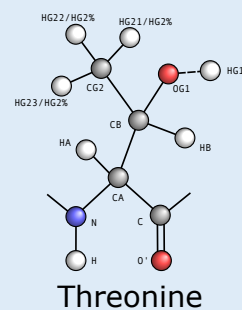
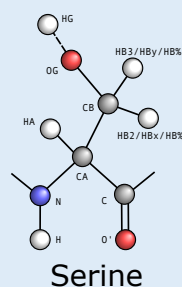
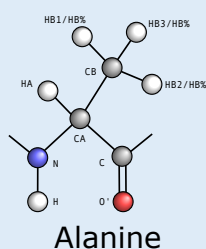
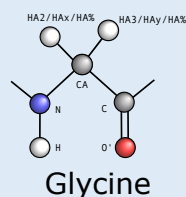
- You can deassign the peak by **right-clicking** on it and going to **Deassign Peak(s)**.
- Then reassign the peak by **dragging NmrAtoms/NmrResidues** onto it or use the **Peak Assigner** to reassign it correctly.

Note: Dragging NmrAtoms or NmrResidues onto a Peak which has already been assigned will add additional assignments to the peak, not replace them.

Carbon chemical shifts for the 20 natural amino acids



20 natural amino acid structures with NEF atom names



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